

LIVERPOOL JOHN MOORES UNIVERSITY

A PRIVACY-PRESERVING BLOCKCHAIN DRIVEN FEDERATED LEARNING FOR HEALTHCARE
DATA AND IMAGE ANALYSIS

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Final Report

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DEDICATION

First of all, thank you God Almighty for everything in my life. I dedicate my thesis work to my beloved wife, Elizabeth Janice George whose love for me knows no bounds and my wonderful baby daughter, Nila Mariam Rijo, who have been around the corner since I started this research. A special thanks to my wife whose words of encouragement and push for tenacity amidst the worst case of job loss during this pandemic situation held me together to complete my research. Her perseverance of taking care of our daughter and not letting the kid interrupt my work despite her work schedule is commendable. Thank you for being the strong pillar to pursue my dreams.

I also dedicate this work to my late father, T. J. Thomas and my late brother, Roy Thomas. Even though they are gone forever away, their memories continue to fill the void they left in our lives. A special feeling of gratitude to my mother, Reena Thomas who have always loved me unconditionally and who taught me the value of hard work. Thank you, my big brother, Razick Thomas who helped in all things great and small especially during my financial crisis and my younger brother, Roji Thomas who has been a constant source of support throughout.

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ABSTRACT

Artificial Intelligence (AI) and Blockchain plays a tremendous role in the healthcare domain, in which AI exceptionally aids doctors to detect, predict and diagnose diseases with excellent accuracy and cost-efficiency. AI dispenses knowledge databases by storing and processing enormous data that facilitates superior disease prediction and thus enhancing clinical decision support. An efficient way to collaborate with the healthcare organizations and to access and share data without compromising security and confidentiality of patients is the need of the hour. To overcome the challenges of sharing sensitive patient information for AI development, federated machine learning (FL) can be leveraged to an extent. Federated learning is a distributed and decentralized machine learning technique to collaboratively train the model without exposing the real data, instead they share model gradient parameters. However, studies emphasis that conventional FL opens gaps in terms of privacy leakage that needs immediate attention. Blockchain technology is a propitious solution to the aforementioned security and privacy concerns due to its immutable, decentralized and trustless behavior. This research investigates on combining Blockchain and Federated learning to address the underlying privacy issues. Even though, numerous privacy-preserving techniques have been proposed, these methods come with trade-offs such as communication outlays and reduced model accuracy. This paper proposed a Blockchain driven FL framework implemented in a cloud based Trusted Execution Environment (TEE) with an efficient encryption-decryption mechanism without any above-mentioned downsides. Finally, the results demonstrate the robustness and enhanced privacy preserving features of the framework and the model accuracies as compared to traditional differential privacy methods. The research concludes that as the TEE is evolving in the cloud environment, it becomes effortless to spin up a TEE enabled server and in future, blockchain technology along with GPU virtual machines supported cloud TEE will facilitate secure and better AI development.

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LIST OF ABBREVIATIONS

AI.....	Artificial Intelligence
BC.....	Blockchain
EHR.....	Electronic Health Record
FL.....	Federated Learning
PBFT.....	Practical Byzantine Fault Tolerance
PoW.....	Proof of Work
PoS.....	Proof of Stake
HIPAA.....	Health Insurance Privacy and Protection Act
GDPR.....	The European General Data Protection Regulation
HL7.....	Health Level Seven
FHIR.....	Fast Health Interoperability Resources
IPFS.....	InterPlanetary File System
DICOM.....	Digital Imaging and Communications in Medicine
BFL / BCFL...	Blockchain based Federated Learning
PDC.....	Private Data Collection
AWS.....	Amazon Web Service
TEE.....	Trusted Execution Environment
MPC.....	Secure Multi-Party Computation
HE.....	Homomorphic Encryption
SSS.....	Shamir's Secret Sharing

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CHAPTER 1: INTRODUCTION

1.1 Background

Artificial Intelligence (AI) is a disruptive technology that has the potential to transform healthcare industry and is poised to be the driving factor that aids to gain unprecedented insights about disease diagnostics and patient care spectrum. The application of AI in medical imaging has opened a new era to facilitate detection of even small abnormalities and other extensive clinical conditions with impressive accuracy, leveraging AI techniques such as computer vision, traditional machine learning and deep learning (Kaul, Enslin and Gross, 2020).

The healthcare domain is a sensitive-data intensive sector and always challenge-driven. Access control, data integrity, provenance and interoperability are critical to maintain both patient's privacy and data exchange between disparate systems (Hasselgren *et al.*, 2020). For the medical AI applications to offer valid, accurate and effective results, larger amount of imaging and clinical data is required. Even though, the world's patient databases contain enough data to acquire promising outcomes, the inadequacy of systematized electronic medical records across different medical facilities poses a significant barrier to data availability for constructive algorithm training. Therefore, to achieve desired health results, collaboration between multidisciplinary health teams is imperative, as shown in Figure 1.1 and a secure medical data exchange is crucial in this regard.

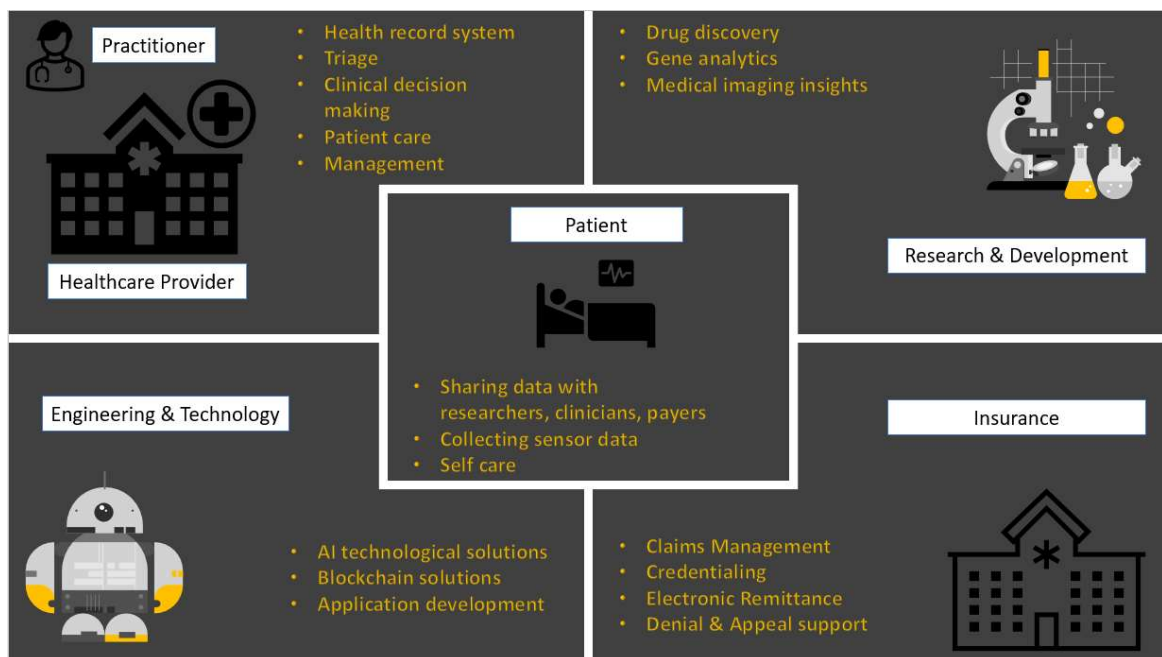


Figure 1.1. Collaborations in healthcare

Moreover, stringent laws and regulations such as HIPAA and GDPR to protect patient privacy still poses a security concern to access these data which causes a significant hindrance to AI development (Kaissis *et al.*, 2020) (Liddell, Simon and Lucassen, 2021). As the advent of Health Level Seven (HL7) Fast Health Interoperability Resources (FHIR) Release 4, a new standard for better interoperability to exchange health data can be secured (Ayaz *et al.*, 2021a). Even though with the implementation of FHIR standards, the current medical systems destitute transparency, provenance, trust, security, and privacy (Yaqoob *et al.*, 2021). Apart from the privacy and security concerns, conventional ML frameworks require data transferred to a centralized server for ML/DL model training which indeed demands heavy communication resources.

A secure and privacy-preserving AI approach can be achieved by employing decentralized data and federated machine learning (FL). FL is an encrypted privacy-preserving decentralized collaborative machine learning framework proposed by Google where the data possessors collectively train the model and in this process none of the data owners expose its real data, instead share only the training gradients. This technique enables the training of algorithms on data while the data remain with its owner (Yang, Liu, Cheng, *et al.*, 2019). Thus, FL shuns the need of transferring the data to the central server for model training, thereby mitigating the risk of privacy leakages and transmission cost. Nevertheless, traditional federated learning comes with security and privacy risks such as central server failure, adversarial attack, privacy leaks and lack of motivation (L. Li *et al.*, 2020a). Even though, various methods have been devised to eliminate these issues, it falls short of communication latency and computational performance.

1.2 Problem Statement

1.2.1 Privacy leakage concerns in Blockchain based federated machine learning

The federated learning approaches combined with blockchain technology have arguably turned out to be the forthcoming privacy preservation technique in medical AI applications. In blockchain based FL, blockchain is used to store only the local and global model gradients and parameters, but not the entire data (R Kumar *et al.*, 2021). (Aich *et al.*, 2021) proposed a BCFL solution to protect personal healthcare record where blockchain is used to store and verify the accessibility of data, which is stored in an external database, while (Passerat-Palmbach *et al.*, 2019) introduced an architecture for federated learning in healthcare consortia in Ethereum blockchain. However, federated learning is not invincible to security and privacy threats. Back deduction while the transmission of gradient parameters has an adverse effect on patient's confidentiality (Z. Wang *et al.*, 2019a). Thus, FL is not adequately secure. A detailed overview of vulnerabilities is shown in Fig 1.2. The general attack types are categorized into Data poisoning attack, Model poisoning attack and Inferring attack (L. Li *et al.*, 2020a) (Zhang *et al.*, 2020).

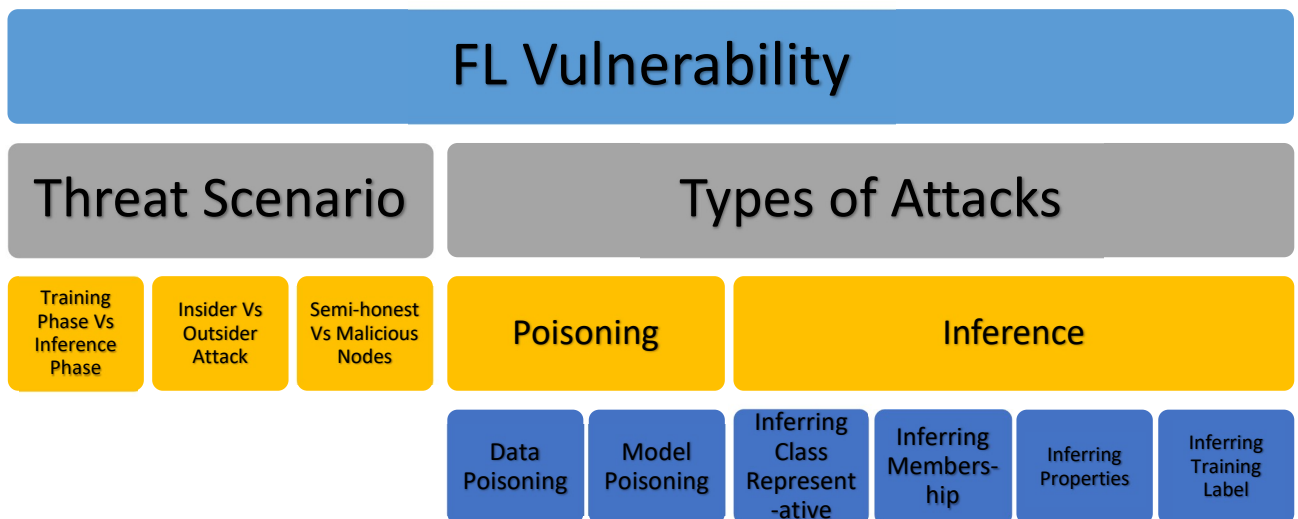


Figure 1.2. An overview of FL Vulnerabilities. Source (Jatain, Singh and Dahiya, 2021)

Numerous blockchain based federated learning techniques have been proposed to overcome existing issues in tradition FL such as single-point failure, poison attack, lack of motivation and inference attacks. (Shayan *et al.*, 2021a) has brought forward a solution to provide privacy-

preserving P2P FL through a distributed blockchain ledger, eliminating single point failure, poison attack and privacy leakage. FLChain (Bao *et al.*, 2019a) is a BFL platform with trust and a reward mechanism to motivate the trainer to be honest and to punish the dishonest node. An incentive is accorded to the nodes based on the size and quality of data samples that affects the accuracy of FL (Kim, Park, Bennis and S. Kim, 2020). DeepChain (Weng *et al.*, 2019a) achieved three security goals – confidentiality for privacy of local gradients, auditability and fairness. (Kang *et al.*, 2019a) stated that a contract-theory based incentive scheme in a consortium blockchain can entice high-regarded clients those have better model accuracy with superior-quality local training data to secure valid FL, thereby removing compromised and malicious workers.

However, as mentioned in the SLA (Hou *et al.*, 2021a) to prevent privacy leakages through inference attacks, BFL architectures must employ additional privacy-preserving methods that lead to communication overheads, efficiency-security trade-off and computational performance. Hence, even within the blockchain technology, FL still poses a gap between confidential data protection and data availability for AI developments. This research aims to compose a healthcare consortium blockchain driven FL framework to mitigate all types of security and privacy vulnerabilities, specifically privacy leakages due to inference attacks without utilizing auxiliary privacy-preserving techniques.

1.3 Research Question

The following research questions are suggested for each of the research objective as highlighted as follows.

1. How can a healthcare consortium blockchain reinforce the security of medical information in a distributed Federated learning environment?
2. How to resolve potential privacy issues in a Blockchain based FL without impacting computational performance and model accuracy-security trade-offs?

1.4 Aims and Objectives

The main aim of this research is to propose a permissioned blockchain driven federated learning framework for healthcare data and image analysis that mitigates all possible security and privacy breaches of highly sensitive healthcare data without a trade-off in terms of performance and accuracy of the final model.

The research objectives are formulated based on the aim of this study, which are as follows

- To implement a healthcare consortium blockchain network in Hyperledger Fabric where the clients of FL are the nodes of blockchain who are the participants in the local model training as well as the global model aggregation.
- To investigate on how to store local and global model gradient parameters securely by leveraging the private data collections (PDC) and key level endorsement policy features of Hyperledger Fabric.
- To investigate and devise a privacy preserving technique without compromising computational performance and accuracy of the model.
- To examine the security and privacy aspects of the implemented BCFL framework.

1.5 Significance of the Study

This study will provide new insights into decentralization of federated learning process to overcome the risks inherent even in blockchain driven FL such as privacy infringements. Mitigating these risks combined with an effective incentive scheme for local model training will attract more organizations, research institutions and healthcare providers to form a healthcare consortium. As the consortium expands, data become large. More data leads to efficient AI developments. As a consequence, it unlocks new business avenues and eventually improve patient care, ranging from better medical image analysis, over precision medicine, to less cost and time drug discovery.

Moreover, the studies, analyses and demonstrations presented in this research will push for new paradigms which will be useful for further studies to innovate current strategies being employed in healthcare blockchain.

1.6 Scope of the Study

The study primarily focuses on developing a healthcare consortium blockchain to securely share model gradients for federated learning. For the study purpose and simplicity, the consortium will be confined to maximum 3 organizations. Technically, the research will be carried out in only one AWS account with all the organizations set up in the different clusters. As the study aims to investigate the security and privacy preservation of data in federated learning, accuracy improvement of FL results will not be considered. Hence, this research will attempt to utilize an existing FL method. A sample dataset of COVID-19 taken from Kaggle will be distributed equally among the three organizations to apply FL local training.

1.7 Structure of the Study

Chapter One sketches the outline of the research. This chapter explicates the research background, and lays the groundworks for choosing the research subject. In addition, the first chapter includes explanation of the research aim and objectives, and describes research structure.

Chapter Two represents the literature review and synthesizes conceptual models and theoretical frameworks that have been previously introduced to the research study. This chapter provides an overview of areas in which the research is diverse and multi-disciplinary. Moreover, this section synthesizes research findings to explore areas in which more research is needed. The research problem specifically has been presented in a systematic and transparent manner in this chapter.

Chapter Three addresses research methodology. The chapter touches on the research process and formulates the path to be used in conducting the research. Moreover, methodology chapter justifies the research design selections, explains why the chosen approaches and techniques are ideal for the research aims and objectives with scientifically sound findings.

Chapter Four focuses on implementation and experimental setup for this research. In addition, the implementation cost is also calculated and shown in this chapter.

Chapter Five demonstrates the results of the experiments conducted in this research and **Chapter Six** concludes the research work by discussing key contributions and future scope of the research.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Data is the centre for AI applications. Deep learning techniques are capable of achieving better accuracy than conventional ML algorithms. However, to pull off excellent model accuracy, massive amount of data must be provided to deep learning models. There are numerous data-sharing efforts centered on genetic sequences, genomic analyses, medical imaging, patient clinical information etc., spanning the worldwide, national and local levels for efficient AI developments (Clark *et al.*, 2013; Wang, Jensen and Zenklusen, 2016; Sayers *et al.*, 2021). Nevertheless, these data sharing efforts for conventional standalone deep learning settings often pose extensively high computational setbacks. This computational overhead problem, however, can be resolved by making use of distributed deep learning techniques. Unfortunately, privacy and security issues are the primary concerns in the distributed deep learning environment and extensive research have been underway in recent years to solve this issue.

Various privacy-preserving deep learning models have been around lately (Chen and Zhong, 2009; Bansal, Chen and Zhong, 2011; Yuan and Yu, 2012; Shokri and Shmatikov, 2015; Aono *et al.*, 2017a). The inception of Federated Learning (FL) (Yang, Liu, Chen, *et al.*, 2019) has been a game changer to solve the aforementioned problems. FL follows the parameter-server architecture and is a collaborative and distributed learning technique which helps the participants associated with the distributed training process to collaboratively train the model to achieve better accuracy and performance while protecting the data privacy of the organizations. The first step in FL is the local deep learning model training by each participant nodes to calculate the local gradients with their own data and uploads the local gradients to the parameter server. The server aggregates those parameters and updates the global model parameters, after which the updated global gradients are distributed to each node to train on the same local data. This process is repeated until the global model converges.

The federated learning framework, however, is vulnerable to certain failures and attacks. The centralized nature of FL for global model aggregation is not viable as it leads to single point of failure (Korkmaz *et al.*, 2020a). Various research demonstrated that malicious participants can poison the global model by injecting falsified local gradients which leads to the corruption of gradient parameter while updating (Bagdasaryan *et al.*, 2020a). Another type of attack is the

inference or adversarial attacks where the gradients can be used to infer sensitive and private information about the training data since the gradients contain sufficient data features (Song, Ristenpart and Shmatikov, 2017; Orekondy *et al.*, 2018). This research investigates and attempts to implement a secure and decentralized blockchain framework for distributed federated learning which can provide data privacy, security and auditability.

2.2 Artificial Intelligence in Healthcare and its challenges

2.2.1 ML/DL applications in healthcare

The high priority need of the hour for AI healthcare is the diagnostics, management and prediction of diseases based on medical imaging. The healthcare sector has witnessed many fascinating breakthroughs in this area where AI has assisted health professionals to detect early abnormalities and provide more accurate diagnostics for various kinds of diseases. While machine learning techniques have been devised to interpret microarray data to classify and detect abnormalities (Molla *et al.*, 2004; Pham, Wells and Crane, 2006; W Shi *et al.*, 2017), the research paper (Cai, Gao and Zhao, 2020) depicts the usage of deep learning for image recognition in early detection of pathologies in medical imaging such as Alzheimer's disease and diabetic retinopathy, ultrasound detection of breast modules, left ventricular segmentation and gastric cancer.

The sheer strength behind a productive AI-based algorithm is data, which is growing at an expeditious rate due to the exponential growth of EHRs from medical diagnoses, prescriptions, demographics, clinical narratives and lab results to medical images such as X-ray films, MRIs and EKGs (Saravanan *et al.*, 2021). Also, with the estimation of 25 billion connected IoMT devices including wearable devices by 2025, the healthcare sector is going to behold an enormous data (Dwivedi, Mehrotra and Chandra, 2021), which can be availed by healthcare workers and researchers to bring the best out of AI developments.

However, AI-based medical solutions often accompany data sharing challenges due to infrastructural obstacles, social and economic factors and legal constraints which curbs the

accessibility of massive healthcare data. The benefits and barriers associated with data sharing is depicted in Figure 2.1 (Pereira *et al.*, 2021).

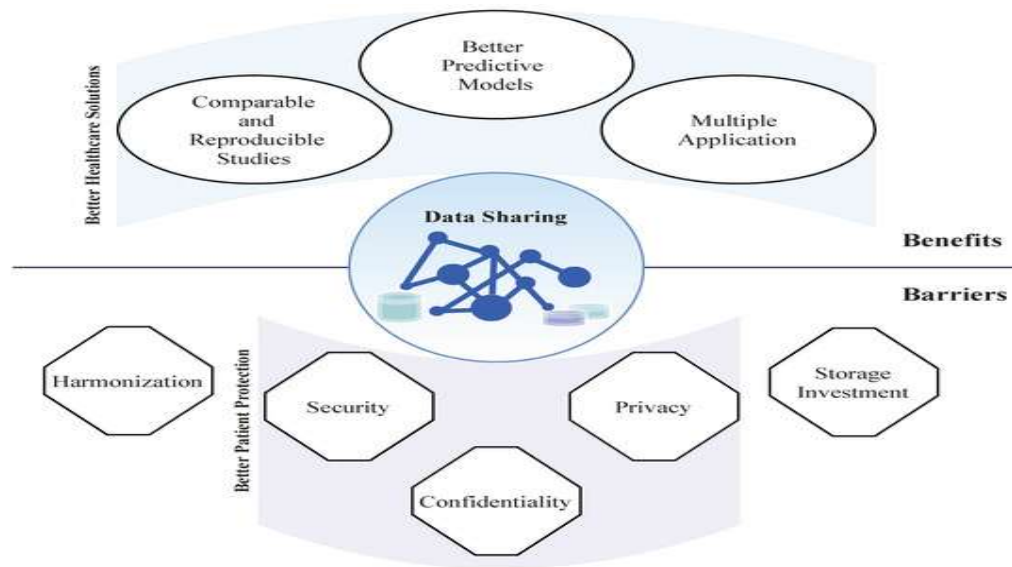


Figure 2.1. Data Sharing Benefits and Barriers (Source Pereira *et al.*, 2021)

2.2.2 The Security and Privacy of Machine Learning/Deep Learning

The most important and rudimentary issues to be addressed in data sharing in the healthcare setting are privacy, confidentiality, and security. There are various research that point out the vulnerabilities of ML/DL models to security and confidential threats like inference attacks (Y. Zhang *et al.*, 2020), adversarial attacks (Biggio, Nelson and Laskov, 2012) and data poisoning attacks (Alfeld, Zhu and Barford, 2016).

2.2.2.1 Security Threats

The security hazards on AI systems can be classified into three types. Table 2.1 shows the taxonomy on the security threats (Qayyum *et al.*, 2021).

- 1) Influence attacks – Modify training data by injecting poisoned data.
- 2) Security violations – Security and privacy violations in the training data.
- 3) Attack specificity – Defines specific objectives of the adversary.

Table 2.1. Security threats in ML/DL

<i>Sl. No</i>	<i>Security Threats in ML/DL</i>	<i>Types</i>	<i>Description</i>
1	Influencing Attacks	Causative Attack	Be in control of the training data
		Exploratory Attack	Exploits the miss-classification of ML models
2	Security Violations	Integrity Attack	Increase the type II error of the model by providing harmful inputs
		Availability Attack	Increase in the type I error of the classifier in response to adversarial inputs
		Privacy Attack	Reveals the sensitive and confidential information of the training data and model
3	Attack's Specificity	Targeted Attack	Attack is directed to a particular input data
		Indiscriminate Attack	Leads to random failure of ML models

Further, three main risks involved in data privacy are (Thapa and Camtepe, 2021)

- 1) Singling out – Identifying features in a dataset by isolating the data
- 2) Linkability – Identifying a feature in a dataset by associating by two or more related features.
- 3) Inference – Identifying the features from different attributes with a significant probability.

2.2.2.2 Adversarial Attacks

Adversarial machine learning attacks are highly critical invasions where the attacker modifies the input to bypass the integrity of the ML/DL models (Szegedy *et al.*, 2013; Goodfellow, Shlens and Szegedy, 2014; Papernot *et al.*, 2016). There are two types of adversarial attacks namely (1) Poisoning attacks where the attacker manipulates the training data to make an impact on the model training (Biggio, Nelson and Laskov, 2012); and (2) Evasion attacks where the integrity of the AI model gets affected by exploiting the test data (Biggio *et al.*, 2013).

Preserving the privacy of the patient involves the security, privacy and trust of personal information (PI) and patient health (PH) data. By any means the model training and inference should not disclose PI and PH data of the patient. Several countries have initiated privacy and

security laws of health data to preserve privacy of the patients. Table 2.2 shows a brief coverage of regulations in the USA (HIPAA), EU (GDPR) and Australia (The Privacy Act) (Thapa and Camtepe, 2021).

Table 2.2. Data Regulations in USA, EU, and Australia

Data Regulation	Requirements
USA – The Health Insurance Portability and Accountability Act (HIPAA)	Proper consent from the patient
	Medical data must be private, auditable and available
	Proper management, physical and technical protection
	Infringement warnings
EU – General Data Protection Regulation (GDPR)	Transparency and Non-discriminate
	Limited data storage and collection
	Safe data processing
	Secure and compliant data sharing
	Rights to patient's data and proper consent
	Infringement warnings
Australia (Privacy Act)	Anonymous yet transparent administration of personal information
	Proper consent and breach notification
	Privacy, security and integrity

2.2.3 Security and Privacy-Preserving techniques in healthcare

As the security and privacy of PI and PH information have been mandated by data laws and regulations, it is imperative to employ privacy-preserving techniques that are applicable to medical data. Various techniques have been researched and implemented such as cryptographic, differential privacy and federated learning to address the security and privacy concerns (Thapa and Camtepe, 2021).

Trusted execution environment (TEE) keeps the confidentiality and probity of data and application-level code by using a hardware or software module which in turn provides a secure

quarantine of confidential computations from other processes. One example of TEE is Intel SGX (Software Guard Extension) which allows to create a reverse sandbox that protects enclaves from operating systems, hypervisors, BIOS, firmware, drivers or any remote attack (Zhao *et al.*, 2016). A use case of Intel SGX in healthcare is the KONFIDO data exchange project (Staffa *et al.*, 2018).

Homomorphic Encryption (HE) is an encryption scheme that addresses privacy concerns by allowing any third-party system to perform computation on the encrypted data without decrypting it while preserving the features and format of the encrypted data (Acar *et al.*, 2018). HE was implemented to protect the privacy and security of genomic data in i2b2 project (Raisaro *et al.*, 2018). However, HE comes with downsides to be used in general applications due to high computational costs and overheads.

Multiparty Computation (MPC) enables all parties to carry out a joint distributed computing task on their own input data such that the parties should learn their output only, nothing else. This scheme follows a secure computation protocol with privacy and correctness and considers the possibility of deliberate malicious activity by some adversarial entity (Lindell, 2020). Even though, MPC incurs low computational cost, it has high communication cost. MPC has to be combined with other privacy preserving techniques to avoid privacy leakages due to collusions.

Differential Privacy (DP) ensures the privacy of the results from computation by allowing to reveal only negligible amount of leakage of input data. Laplace and exponential mechanism are some common algorithms to add noise to the input data to achieve differential privacy (Dwork, 2006). DP assumes the data owners to be trustworthy and there should be a proper trade off between privacy and the quality of data to accurately learn the data. DP combined with MPC is implemented as an underlying privacy-preserving technique in a healthcare project namely SODA.

2.3 Federated Learning

2.3.1 Brief Introduction to FL

Federated learning (FL) is a distributed and privacy preserving machine learning technique in which the model can be trained without a non-trusting party accessing the private data. FL follows a client-server architecture, also called parameter-server architecture where the server is responsible for collaboratively training a global ML model and distributing the global model to the clients who are responsible to train the models on their local data and interchange parameters to update the global model. It was introduced by Google to predict user's text input within numerous mobile devices while keeping data on the devices (McMahan *et al.*, 2017). The FL process generally consist of three key steps which is depicted in the Figure 2.2. Firstly, each participant or device in the federate of training process downloads the initial global model gradients for local training with their own data. Secondly, the updated local model gradients from each device are sent to the central server for aggregation. Finally, the global model gradient is updated with the average of all the local model gradients and dispatched again to the devices. This training process gets recursed until the global model meets an expected threshold value (L. Li *et al.*, 2020b).

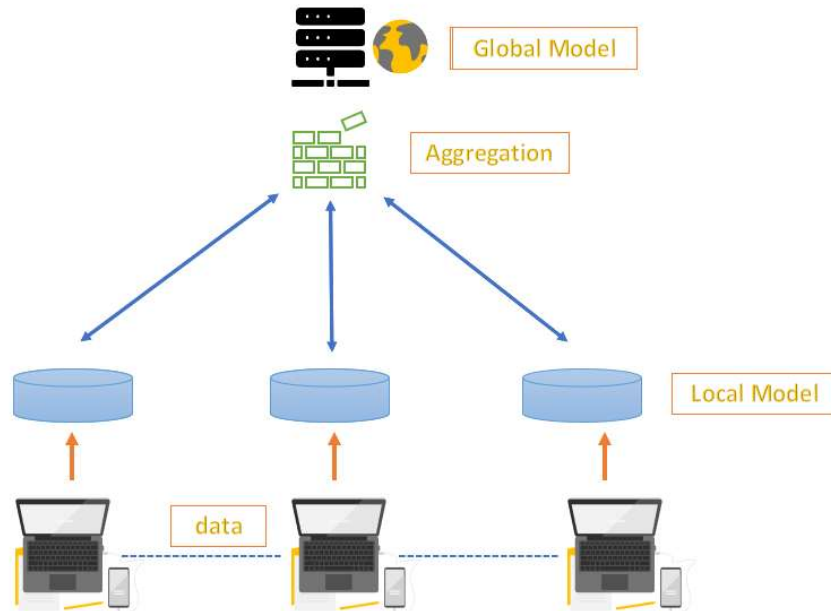


Figure 2.2. FL architecture

Federated learning can be categorized into three types, namely Horizontal, Vertical, and Transfer federated learning. Figure 2.3 depicts a representation of horizontal, vertical and transfer FL. In horizontal FL, the dataset is split horizontally as the user attributes of multiple data sets overlies more than the user itself, whereas in vertical FL, the data is divided in the vertical direction when the user dimension overlap is more than the attribute space. Federated transfer learning is usually used for models with deep neural networks when the data is scarce (Jiang *et al.*, 2020; L. Li *et al.*, 2020b).

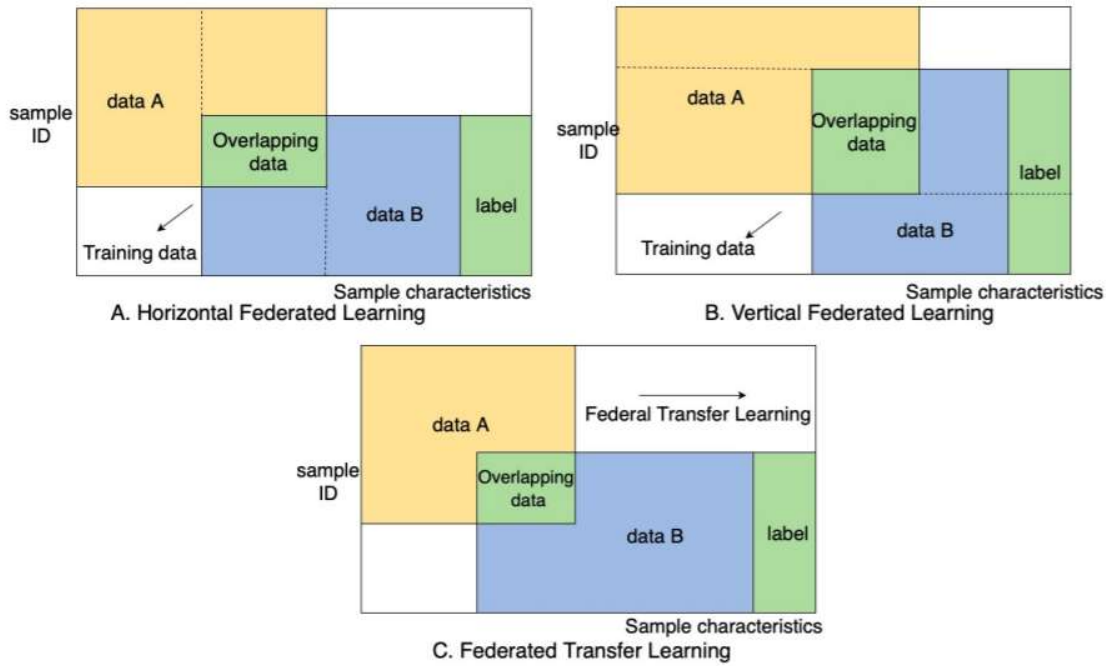


Figure 2.3 Horizontal, Vertical and Transfer FL (Source Jiang *et al.*, 2020)

2.3.2 Privacy Preserving in Federated Learning

There are different privacy-preserving schemes that have been adopted in FL. Table 2.2 shows the summary of common PPFL methods and Figure 2.4 represents a taxonomy to privacy-preserving Federated learning. Privacy-Preserving federated learning (PPFL) can be broadly classified into (Yin, Zhu and Hu, 2021)

1. Encryption-based – Mainly use cryptographic techniques to achieve privacy preservation such as homomorphic encryption (HE) (Y Chen *et al.*, 2020), multiparty computation (MPC) (Bonawitz *et al.*, 2017) and Secret Sharing algorithms (Gao *et al.*, 2019).

2. Perturbation-based – Differential privacy methods are used to defend against reconstruction attacks (Lu *et al.*, 2019).
3. Anonymization-based – Anonymization schemes have been proposed to improve the data utility and model performance while preserving privacy of the data
4. Hybrid – To overcome the computation and communication overheads in encryption-based and perturbation-based schemes, hybrid PPFL have been proposed to balance the trade-off between data utility and data

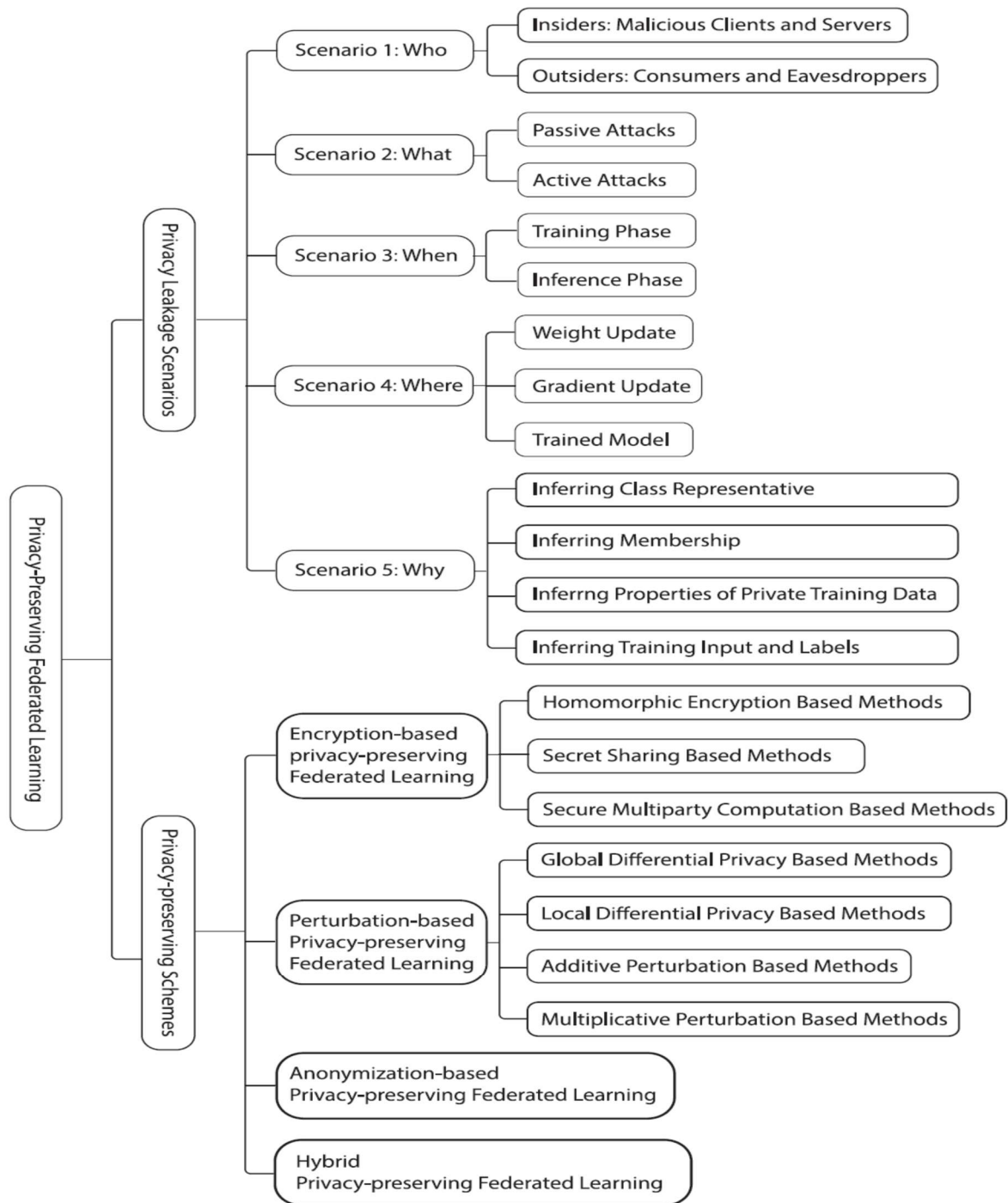


Figure 2.4 Taxonomy to Privacy-Preserving FL (Source Yin, Zhu and Hu, 2021)

Table 2.2 PPFL methods

	Homomorphic Encryption	Multiparty Computation	Differential Privacy
Data protection	Storage & Computing	Computing	Output Data
Output data privacy	No	No	No
Computation overhead	High	High	Low

Communication overhead	Not required	High	Not required
Cryptographic technique	Yes	Yes	No
Verifiability	No	Yes	No
Implementation Complexity	High	High	Very low
Collective Computing	Applicable	Suitable	Suitable
Interactive	No	Yes	No

2.4 Blockchain in Healthcare

The advent of Bitcoin in 2008 was the dawn of Blockchain technology (Nakamoto, 2008). Blockchain is shaping a new revolution in the healthcare eco system with its cryptographic, transparent, immutable, distributed, and trustless nature. By fully utilizing this cutting-edge technology, this decade will witness the evolution of healthcare industry to a patient-centric model with enhanced security, privacy, and interoperability of health data. One of the prominent use cases of blockchain in the healthcare is medical record access and sharing. Blockchain powered healthcare applications can employ patient-controlled accessibility to put together patient records easily, thereby enabling hassle-free exchange of data across different institutions (Hölbl *et al.*, 2018).

There exist three types of blockchain namely public, private and consortium or federated (Morkunas, Paschen and Boon, 2019). Table 2.3. shows the summary of types of blockchain. A consortium based private blockchain is preferable for healthcare use cases as it offers additional degree of transaction privacy, which is critical for sensitive data transactions. Moreover, consensus mechanisms like PBFT facilitate greater transaction throughput, scalability and cost cutting. Since only verified parties are allowed to initiate future participants in the consortium, it adds higher level of trust, reliability and security. Table 2.4. shows an overview of consensus mechanisms.

Table 2.3 Types of blockchain summary (Hasselgren *et al.*, 2020)

Property	Public Blockchain	Consortium Blockchain	Private Blockchain
Consensus Nodes	Miners	Selected nodes	Governance Structure
Transparency	Public	Public or restricted	Public or restricted
Immutability	Nearly impossible	Partial	Partial
Throughput	Low	High	High
Energy Efficiency	Low	High	High
Centralized	No	Partial	Yes

Authentication	Permissionless	Permissioned	Permissioned
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Table 2.4 Types of consensus algorithms overview (Xiao *et al.*, 2020)

Property	Proof of Work (PoW)	Proof of Stake (PoS)	PBFT
Structure	Permissionless	Permissionless	Permissioned
Energy consumption	High	Medium	Low
Adversarial hashing power toleration	< 25% computing	< 51% stake	< 33.3% faulty replicas
Transactions per sec	7 - 30	30 - 175	Upto 20000
Example	Bitcoin	Dash	Hyperledger Fabric

2.5 Blockchain Based Federated Learning

Several research manifest the existing issues in the conventional federated learning such as single point failure, lack of motivation, poison attacks and privacy leakages. Blockchain technology with its immutable and verifiable property has great potential to build a secure FL framework in an untrustful environment. Some existing blockchain based federated learning architectures have addressed the abovementioned issues and is summarized below.

2.5.1 Single Point Failure

In conventional Federated learning, models are vulnerable to server breakdowns and external invasions. Third party blockchain networks have been applied to FL to address the single point failure concerns (Kang *et al.*, 2019b, 2020; Weng *et al.*, 2019b; Korkmaz *et al.*, 2020a; K. Zhang *et al.*, 2020; Ma *et al.*, 2020; Shayan *et al.*, 2020; Wu *et al.*, 2020; Y. Li *et al.*, 2020; Chen *et al.*, 2021; Desai, Ozdayi and Kantarcioglu, 2021; Peng *et al.*, 2021). (Majeed and Hong, 2019) devised a blockchain based federal learning architecture (FLchain) where the blockchain architecture is inspired from both public and private blockchains (Ethereum and Hyperledger) and the network is formed of edge devices. They utilized the channel feature of Hyperledger Fabric to train each global model in the network and securely stored the global model as Merkle tree. ChainFL uses a private Ethereum based blockchain architecture to upload the global model to the miners in the network. Since the miners run the smart contract deployed on the network to aggregate the global model from each node, this system alleviates single point of failure. Moreover, any participants can access the immutable history of the global model as it is stored in the blockchain (Korkmaz *et al.*, 2020b).

In the paper (Li *et al.*, 2021), BLADE-FL is a trusted public blockchain system that comprises of clients with equal computing power. Each client has the role of trainer as well as the miner where the trainer trains the model and broadcast it to the network. The miner mines the block that includes all the local models and then each client performs the aggregation of global model, assuming that the model cannot be tampered with the external attackers. The works (Qu *et al.*, 2021; Uddin *et al.*, 2021) also shows how the model aggregation by miners in a fully distributed blockchain network can resolve the single point failure issue.

Even though the above researches resolve the single point failure problem, the solutions often comes at the expense of heavy energy consumption for mining and extra mining latency. Those also don't address other issues such as lack of motivation, poison attacks and privacy leakages.

2.5.2 Lack of Motivation

BlockFL (Kim, Park, Bennis and S.-L. Kim, 2020) proposed a blockchain federated learning architecture to overcome the single point of failure problem and motivates the federation by providing rewards proportional to the training data. They also addressed the delay due to the blockchain network by proposing an end-to-end latency model by considering computation, communication and Proof-of-Work delays. On the other hand, FLChain (Bao *et al.*, 2019b) uploads cryptographically masked gradients to the blockchain. The reliability of each client trainer will increase based on partner's and user's evaluation and thus a price is set for the client's model. The client who detects the misbehavior will be rewarded, dishonest behavior will be penalized, and the affected client will be compensated by the FLChain network.

In (Ma *et al.*, 2020), a blockchain FL was developed inspired from BLADE-FL (Li *et al.*, 2021) with an additional reward strategy where the task publisher allocates rewards to the participating nodes proportional to the size and quality of the training data.

A consortium blockchain can be used to provide incentives and misconduct deterrence (Kang *et al.*, 2019b) by introducing poison attack detection schemes like RONI (Barreno *et al.*, 2010) and unreliable workers schemes like FoolsGold (Fung, Yoon and Beschastnikh, 2018). The attacker detection and unreliable worker schemes ensure the task publisher to remove unreliable and malicious local model updates from unreliable workers and poison attackers respectively. Moreover, the honest workers obtain incentives from the task publishers according to their

resource contribution and model training behaviors. Finally, the miners add the reputation opinions as a block and add the block in the reputation consortium blockchain after block validation using a PBFT consensus mechanism. (Desai, Ozdayi and Kantarcioglu, 2021) proposed a blockchain based FL framework with an objective to deter adversarial attacks especially backdoor attacks (targeted attacks) and model poisoning using a novel attacker detection algorithm. They leveraged a private blockchain framework (Hyperledger Fabric) to perform the global model aggregation and a public blockchain (Ethereum) to store the cryptographic hashes of parameter set sent to private blockchain and to reward honest worker nodes with a cryptocurrency.

All the above works did an excellent job to encourage the participants in the network to bring their best behaviors through effective incentive mechanisms. Nevertheless, they suffer poison attacks and privacy leakages from malicious adversaries internally (participating clients) and externally (model consumers and eavesdroppers).

2.5.3 Poison Attack

In order to eliminate dishonest behavior, (K. Zhang *et al.*, 2020b) introduced a new evaluation mechanism to evaluate the participants fairly and also proposed an algorithm namely weight-descending candidate selection algorithm for a better participant selection. With the help of Proof-of-Verification (PoV) consensus mechanism (Kang *et al.*, 2020b), the proposed BFEL framework can defend against poison attacks and ensure security due to the immutable, decentralized and traceable nature of blockchain. In addition to that, the malicious miners will be penalized, and their stake will be confiscated which deters the malicious behavior of miners.

(Peng *et al.*, 2021) represents the first attempt to develop a blockchain based FL framework namely VFChain which adheres with verifiability and auditability. A committee is chosen to aggregate the global models and store verifiable proofs in a Hyperledger Fabric blockchain with a novel idea of authenticated data structure algorithm (Dual Skip Chain) to improve the search effectiveness of verifiable proofs. A digital asset called VFCoin is also introduced in the network to reward the workers and committee trustees. BFLC (Y. Li *et al.*, 2020) also addresses these security challenges by taking advantage of an efficient and secure committee consensus mechanism by validating the local updates with committee member's data distribution. Tokens

will be attached to the nodes as rewards for eligible local updates. A validation mechanism inspired by Proof-of-Stake (PoS) consensus mechanism was devised (Chen *et al.*, 2021) to create blocks by the most contributing clients. The block comprises of local model updates along with their corresponding vote results.

All these frameworks are set up under the pre-condition and assumption that the participants are at least semi-honest adversaries and will not collude with each other. That means, at least n trustees are honest and assume to disseminate correct outputs into the network such that n is larger than $2/3$. In some cases, each participant must have equivalent computational resources. Also, these studies not guarantee sufficient privacy preservations.

2.5.4 Privacy Leakages

(Weng *et al.*, 2019b) presents a framework called DeepChain for auditable and privacy preserving federated learning. They proposed a collaborative training blockchain framework with a reward approach that encourages participants to collectively engage in FL model training and share the local gradients by employing Threshold Paillier algorithm to ensure the privacy of local gradients and guarantees auditability of the training process. FedBC (Wu *et al.*, 2020) uses a ring topology approach to determine the node position in the training process where the topology is generated by the smart contract in the blockchain. The node uses the homomorphic public key of the direct successor node to encrypt the changed local parameter value and stores it into IPFS to obtain a hash file index so that the successor node in the ring can decrypt the value with its private key.

Biscotti (Shayan *et al.*, 2020) uses proof-of-federation (PoF) combined with verifiable random functions (VRFs) for hashing to prevent peer collusion. Their approach leverages Multi-Krum defense to prevent model poisoning, achieves privacy preservation through differentially private noise and uses Shamir secrets for secure aggregation. A blockchain based hierarchical crowdsourcing FL system was proposed by (Zhao *et al.*, 2020a) to optimize the functionalities of IoT home appliances. They leveraged differential privacy mechanism to prevent adversaries from exploiting the private and sensitive information of the customers and the blockchain ensured prevention of malicious model updates.

To achieve data privacy and provenance protection goals, (Awan *et al.*, 2019) suggested a Ethereum based PPFL framework where they developed a variant of Paillier cryptosystem to implement homomorphic encryption and proxy re-encryption for gradient aggregation.

The studies conclude that even though Blockchain technology improves the security of the BFL but lacks inherent privacy protection. Therefore, BFL needs to integrate additional privacy-preserving methods to guarantee sufficient data confidentiality. For example, differential privacy can prevent leakage of sensitive information to malicious adversaries by extracting the learned model (Arachchige *et al.*, 2020; Zhao *et al.*, 2020b). Homomorphic encryption, such as Paillier algorithm (Aono *et al.*, 2017b) and Learning with Errors (Paillier, 1999), is employed in BFL to encrypt the training data and prevent the attackers from inferring privacy information. However, these extra privacies preserving mechanism may incur communication and computation overheads and implementation complexity. Moreover, some methods are set up under the assumption of at least n participants are honest and do not collude with each other.

Another work by (Yu Chen *et al.*, 2020) proposed a privacy-preserving FL scheme based on trusted execution environment that ensured privacy and integrity of machine learning training processes. (Rahman *et al.*, 2020) proposed a secure and provenance federated learning to classify IoHT data. They used blockchain and off-chain for training data and models to provide provenance, integrity, and tamper-resistance. Intel SGX TEE provided an additional layer of security for local model aggregation and MPC to reduce communication and computational overheads. StartFL is a hybrid FL architecture aimed to resolve the shortcomings posed by federated learning especially concerns over secret key distribution, output verifications and integrity of client models. They used a combination of TEE and quantum key distribution to mitigate these problems.

2.6 Discussion

Various researchers have demonstrated the integration of blockchain technology with FL to eliminate security risk such as poisoning attacks by utilizing consensus algorithms. Even though, the works mitigated the security concerns, yet couldn't completely take the edge off the security issues. In public blockchains, Proof-of-Work (PoW) is predominantly used consensus protocol, however a threat like 51% attack is still possible for the hostile to take control of the network and cause catastrophe (Gemeliarana and Sari, 2018). In addition, the

computational cost and energy consumption is too high making it a non-viable solution to be integrated with the FL. Some research papers introduced other consensus mechanisms such as PBFT (Castro and Liskov, 2002), PoF (Shayan *et al.*, 2020) etc., however, those too are not invincible to security attacks. For example, PBFT works with an assumption that only one third of the nodes should be faulty or malicious to successfully reach a consensus. In a healthcare setting, a private consortium permissioned blockchain would be pertinent and this research intends to adopt the same.

As we can see in this section of literature review, blockchain technology alleviates the security of data in a FL environment, nonetheless blockchain natively doesn't eventuate in privacy protection. Hence, all proposed architectures must amalgamate additional privacy-preserving techniques such as differential privacy (Dwork, 2006), secure multi-party computation (Lindell, 2020) and homomorphic encryption (Aono *et al.*, 2017a) to prevent malevolent attackers from inferring sensitive data from the models. Anyhow, these mechanisms incur high communication and computational overheads and intricate realization. This research also investigates on how to overcome the above-mentioned shortcomings.

2.7 Summary

Recently, blockchain has been one of the hottest areas of research to address the shortcomings faced by the conventional FL. By taking advantage of the distributed peer-to-peer property of blockchain, single point of failure can be alleviated where the aggregation of global model would occur in all the nodes based on some consensus mechanism (Ramanan and Nakayama, 2020). Further, a consortium blockchain allows only trusted parties in the network, thus avoiding malicious local model updates to participate in the global model aggregation process (Kim and Hong, 2019). Moreover, a reward system can be coined to encourage honest behavior

from all participating clients (Liu *et al.*, 2020). While Blockchain driven federated learning has shown great plausibility in addressing these threats, inferring attacks by applying back deduction is still possible even in blockchain to reconstruct the patient information which is a clear case of privacy breach (Z. Wang *et al.*, 2019a).

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

This research envisages to develop a methodology, through theoretical research as well as a PoC implementation of the proposed architecture. The project will be advanced in 4 phases to reach the objectives as indicated in Fig 3.1: (1) Developing a healthcare consortium blockchain framework; (2) Implementing a novel privacy-preserving mechanism to protect sensitive

healthcare data computation; (3) Applying Federated learning to COVID-19 chest X-ray images; (4) Security analysis of private data.

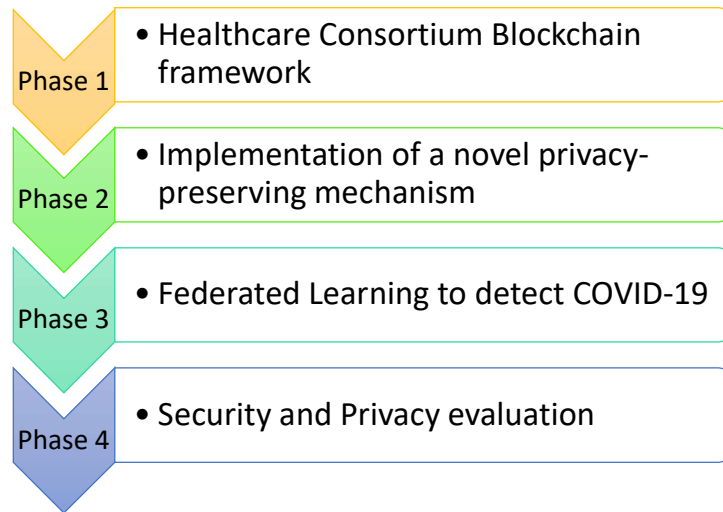


Figure 3.1. Research Phases

The first phase of this research will involve the investigation and setup of a healthcare consortium blockchain in a cloud platform to store the model gradients in a distributed environment where every organizations in the consortium will have the chance to participate in the local training model and global model aggregation. In the second phase, a thorough study on various privacy-preserving techniques is conducted and based on the study, a novel privacy-preserving mechanism will be designed and employed to protect sensitive health data during model training and consuming. The third phase of this research mainly focuses on devising a federated learning approach to detect COVID-19 using a data sample. The final phase will observe a detailed security and privacy interpretation and evaluation of the devised threat model and attack conditions.

3.2 Research Methodology

3.2.1 Problem Definition and Motivation

For an AI model to achieve exceptional outcomes in healthcare, i.e., from disease detection, precision medicine to drug discovery, enormous data is required to train the model. However, healthcare institutions in solitude often doesn't accommodate sufficient data to build an efficient AI algorithm. As a matter of fact, the healthcare ventures must collaborate in a multi-

party setting where they could share the data with a centralized server to train the data to generate robust AI models (Watcharapichat *et al.*, 2016; Xu *et al.*, 2021). Needless to say, such sharing comes with privacy challenges as the health data contains potentially sensitive information. To avoid direct sharing of sensitive and confidential data, Google proposed federated learning (McMahan *et al.*, 2017) and is a promising solution for securing large-scale multi-party ML/DL computations where it advocates sharing of the model parameters and gradients, but not the raw data.

Since all the participants in the federal distributed learning environment is actively participating in the training process, it presents new challenges with regards to security and privacy of sensitive data (Fung, Yoon and Beschastnikh, 2020), for example, patient's health record. Several research studies (Biggio, Nelson and Laskov, 2012; C. Xie *et al.*, 2019; Bagdasaryan *et al.*, 2020b; Fung, Yoon and Beschastnikh, 2020) have demonstrated how malicious actors would attack the shared model through poisoning attacks in which the adversaries inject harmful updates to the shared model parameters. Information leakage is also another concern where the attackers attempt to infer features of the confidential training data by observing the target's learned model (Hitaj, Ateniese and Perez-Cruz, 2017; Melis *et al.*, 2019). Blockchain technology through its consensus mechanisms can avoid poison attacks to some extent, yet consensus algorithms exhibit security concerns such as the 51% or sybil attacks in PoW (Gemeliarana and Sari, 2018) and collusion of more than two third of nodes in PBFT (Castro and Liskov, 2002). Privacy infringements can be defended by differential privacy (Dwork, 2006) or secure aggregation with multiparty computation (Lindell, 2020) and homomorphic encryption (Aono *et al.*, 2017b). In fact, blockchain doesn't offer indigenous privacy preserving mechanism and BFLs must integrate additional privacy-preserving techniques which lead to communication latency, computational overheads, heterogenous issues and implementation complexity (Nguyen *et al.*, 2021).

Due to the security concerns over the sensitive information while sharing the data, the healthcare industry is not yet ready to embrace federated learning which has huge potential for efficient AI applications. The primary motivation for this research is to contribute a secure and privacy preserving technique with reduced communication overheads, computational costs and uncomplex implementation. Also, motivated by the unique properties of blockchain, this research envisages to implement a simple and direct PPFL in a highly distributed P2P blockchain healthcare setting, assuming all the participants in the distributed network evince dishonest behavior.

3.2.2 Theoretical Framework

Inspired by existing trusted execution environment based (TEE) integrated blockchain-FL architectures mainly (Rahman *et al.*, 2020; Yu Chen *et al.*, 2020; Huang *et al.*, 2021), this research's vision is to propose a novel Hybrid blockchain driven federated learning architecture by consolidating Trusted Execution Environment (TEE) to provide an isolated secure computing environment, Secure Multi-Party Computation (MPC) and Shamir's Secret Sharing algorithms for training-integrity and privacy-preserving model aggregation. Theoretically, the operational process of the proposed Blockchain integrated FL constitutes of the following steps. Figure 3.2 represents the theoretical framework.

- The aggregator starts their enclave and sends back the aggregator's TEE enclave public key to the parent instance.
- The parent instance sends the aggregator's enclave public key to the public ledger of the private blockchain.
- Each participant starts their enclave and sends their enclave public key to the aggregator-participant shared private ledger of the blockchain.
- In each participant's parent machine, their local model undergoes training with the initial global gradients and the local data to update local gradient which is then encrypted with the aggregator's enclave public key taken from the public ledger of the blockchain.
- Each participant broadcasts their encrypted local gradients to the shared private ledger of the blockchain.
- After the completion of local training by all the participants, each participant's encrypted local gradient is transferred from the blockchain to the aggregator TEE enclave.

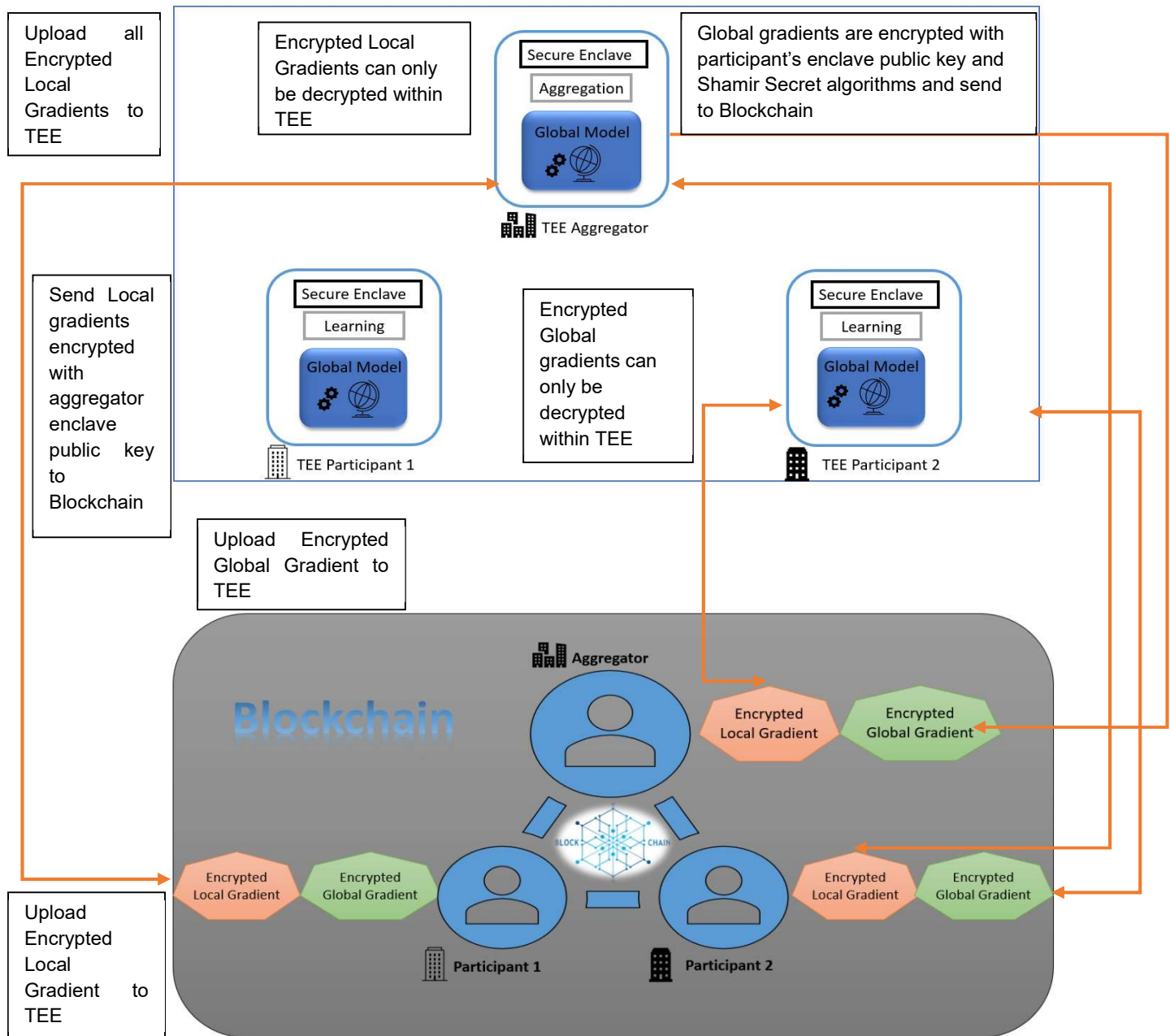


Figure 3.2. Theoretical FL framework

- Within the aggregator enclave, all the local gradients are decrypted, and by aggregating all the updates from the participants, the aggregator constructs new global gradients. The secure enclave helps to decrypt the local gradients without the need of a trusted party.

- The global gradient is encrypted in two ways, one with Shamir's Secret Sharing algorithm and the other with each participant's enclave public key.
- The global gradient encrypted with participant's enclave public key is stored in the respective participant's private ledger and SSS encrypted gradient is broadcasted to the public ledger of the blockchain.

3.2.3 Requirements Analysis

3.2.3.1 Hardware Requirements

For experiments to be carried out in a distributed setting, the following hardware resources are required

- Intel(R) Core(TM) i7 CPU, 8 GB RAM, 64-bit Ubuntu System
- Amazon Web Service (AWS) cloud platform
- AWS Elastic Kubernetes Service (EKS) Cluster - 3
- AWS EKS Node – 2 vCPU, 8 GB RAM, m5.large – 3 Nos
- AWS Elastic File System (EFS)
- AWS Elastic Compute Cloud (EC2) instance – 4 vCPU, 16 GB RAM, m.xlarge – 3 Nos (Nitro Enclaves)
- AWS Elastic Compute Cloud (EC2) instance – 2 vCPU, 8 GB RAM, m.large – 3 Nos
- AWS Simple Storage Service (S3)
- AWS Key Management Service (KMS)

3.2.3.2 Software Requirements

- Visual Studio Code
- Lens IDE
- JupyterLab / PyCharm
- Tensorflow framework
- Docker
- GoLang
- Nodejs

3.2.3.3 Data Description

In this research, the database of chest X-ray images for COVID-19 positive cases is collected from Kaggle (Chowdhury *et al.*, 2020; Rahman *et al.*, 2021) (2020, March. COVID-19 Radiography Database, Version 4. Retrieved January 12, 2022 from <https://www.kaggle.com/tawsifurrahman/covid19-radiography-database>). The database contains 3616 COVID-19 positive cases along with 10,192 Normal, 6012 Lung Opacity and 1345 Viral Pneumonia images. Figure 3.3 depicts sample images from the database for normal, COVID-19 Pneumonia and Viral Pneumonia CXR images (Chowdhury *et al.*, 2020).



Figure 3.3. COVID-19 Chest X-ray Sample (Source (**Chowdhury *et al.*, 2020**))

3.2.4 Assumptions – Threat Model

3.2.4.1 Design Assumptions

The framework assumes a consortium blockchain in which peers can only join or leave the network over the succeeding approvals. All the clients in the consortium are assumed to be adversaries who are willing to collaborate on AI training without sharing their data. In the blockchain setting, the peers are assumed to receive the latest blockchain state in the case where they re-join the network after going offline. At the time of decrypting the global model with Shamir's Secret algorithm, the blockchain network is assumed to be in an idle state where no new clients can be added during that decryption process. Also, the framework assumes that two third of the clients share their secret keys for SSS decryption. Federated learning will assume stochastic gradient descent (SGD) for model aggregation optimization. For each new training session, the clients are supposed to start a new isolated enclave.

3.2.4.2 Threat Model – Security and Privacy Requirements

1. Privacy Leakage Attacks

In the federated learning framework, the aggregator server and the client server are considered as non-trusted parties, i.e., they can be honest, dishonest, or semi-trusted (honest yet curious) participants. Based on this assumption, a curious insider (clients and aggregator server) or outsider (model consumers and eavesdroppers (L. Wang *et al.*, 2019)) could infer sensitive health information from the data by reconstructing the raw data from the shared model parameters (Z. Wang *et al.*, 2019b). The scope of the research exempts model consumers' threat to the final deployed model. Both passive and active attacks are considered for evaluating security and privacy of the data. Passive attacks refer to observing and learning from the FL system, but not altering the system (Melis *et al.*, 2019) and it is assumed that adversary can't access the model gradients and intermediate training results, but only the query result. Active attacks aim to modify the model updates on the system to extract private information from the dataset (Nasr, Shokri and Houmansadr, 2019). It is also assumed that during the information leakage attacks, the adversaries intend to learn users' features from the local dataset as well as attempt to infer membership and class representatives (Hitaj, Ateniese and Perez-Cruz, 2017) information by inference attacks. For the federated learning to have a high degree of privacy protection, the framework relies on Trusted Execution Environment (TEE) in the setting of service platform where the computations are performed within a trustless territory. Moreover, the architecture leverages Shamir Secret Sharing algorithms and a key management service to encrypt the local and global models in which those can be only decrypted within the TEE with its private key.

2. Poisoning Attacks

This research is conducted under the assumption that adversaries intend to harm the performance of the global model through targeted attacks such as backdoor attacks and gradient-ascent attacks, and untargeted attacks like sign-flipping and same-value attack. Targeted attacks aim to affect the minority of data items in the model while maintaining the performance of the model (Bagdasaryan *et al.*, 2020b) whereas untargeted attacks

inject false value or incorrect subset of data in the local dataset to alter the learning outcome of local clients (Li *et al.*, 2019; Peng, Li and Ling, 2021). A secure ledger configuration using a private consortium blockchain and smart contracts to store encrypted models, local data and TEE docker images are desired to meet these security requirements in a federated learning environment.

3.2.5 Evaluation

This section revisits the two primary goals in the evaluation of the prototype and provides an interpretation of the research objectives. The proposed framework

- 1) Protects the confidentiality of individual's sensitive health and personal data
- 2) Can withstand poisoning attacks.

3.2.5.1 Evaluation Setup

The experiments to be conducted in a distributed setting, a consortium blockchain network namely, Hyperledger Fabric was chosen where the network is set up across 3 AWS regions: Mumbai, North Virginia and Frankfurt. An EKS cluster was created for each of the clients, one aggregator and two clients to deploy Hyperledger Fabric in the Kubernetes setting for higher scalability. For the testing purpose, one EKS worker node, 2 vCPUs, 8 GB RAM m5x.large machine was created for each client to setup 2 peers each. The Trusted Execution Environment (TEE) was realized by leveraging AWS EC2 instances with Nitro Enclave enabled for configuring isolated environments for sensitive data computing. AWS Key Management Service (KMS) was used for TEE and other data (local and global models) encryptions. The experiment was conducted on a real-world open-source and free datasets of COVID-19 from Kaggle was used in this research which contains 3616 real-world COVID-19 affected images, 6012 Lung opacity images, 10192 Normal images and 1345 Viral Pneumonia images. The dataset was divided for each participant as per the Table x and the data was distributed among the three participants entirely in a non-IID fashion.

Category	Training Data (Approx)	Test Data (Approx)	Validation Data (Approx)
COVID-19	843	542	542
Lung Opacity	1402	300	300
Normal	2378	510	510
Viral Pneumonia	314	67	67

The global model learning is evaluated with the help of a two-layer convolutional neural network (CNN) with SGD as the optimization technique by utilizing the Tensorflow

framework. The CNN softmax model uses a local batch size of 10, SGD momentum of 0.75 and learning rate of 0.01. Table 3.1 shows the default parameters used in FL experiments. The evaluation for targeted attacks, untargeted attacks and privacy attacks is conducted in the Blockchain database administrative level where the malicious attacker intends to change the value of encrypted local and global gradients in the Blockchain.

Table 3.1. FL Parameters and Default Value

FL Parameter	Default Value
Learning Rate, η	0.01
Local Batch Size, B	10
Number of Local Epochs, E	10
SGD Momentum, β	0.75
Proportion of Trainers, V	1
Number of Clients	2
Number of Aggregators	1
Number of Peers in each participant	2

3.2.5.2 Information Leakage Interpretation

In the aggregator TEE, the global model is double encrypted with hash of all the decrypted local models combined with the hash of encrypted global model by the aggregator encrypted data key. The local models are encrypted with the client's own keys and decryption can happen only in the respective client's TEE or the aggregator's TEE using its private key in run time. Moreover, the models are disseminated to the blockchain for traceability and immutability. This encryption-decryption mechanism within the TEE in a distributed blockchain setting hampers any effort from the insiders (clients or aggregator) or outsiders (eavesdroppers) to infer sensitive health information from the shared models unless all the participants in the network collude which is practically paradoxical in a private consortium blockchain setup. Similarly, attempts from passive and active attacks to extract private information and inference attacks to infer membership and class representatives are also out of question due to the data protection provided by the security and privacy preserving features mentioned above.

4.1 Proposed Blockchain Based FL Architecture

Inspired by the existing Blockchain federated learning architectures and the theoretical framework from this research, this section introduces a Blockchain-FL architecture as shown in Figure 3.8. The system design of this Blockchain-FL architecture comprises of a consortium

based blockchain namely Hyperledger Fabric and AWS Nitro enclave for sensitive health data computation.

Hyperledger Fabric

Hyperledger Fabric is an open-source enterprise level permissioned consortium based distributed ledger technology hosted by Linux foundation (Androulaki *et al.*, 2018). Fabric has a modular architecture to allow the components like membership services and consensus protocols to be plug and play. Unlike most of the other traditional blockchain platforms that require domain specific languages (DSL) to write smart contracts, Fabric can leverage general purpose languages (GPL) to develop smart contracts, also called chaincodes. The introduction of execute-order-validate architecture in Fabric as shown in Figure 3.4 eliminated non-determinism caused by GPLs.

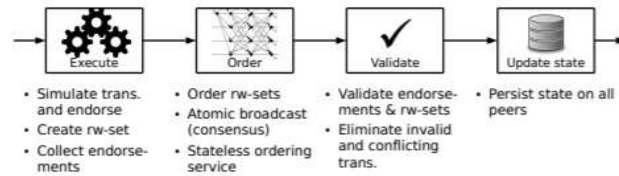


Figure 3.4. Execute-Order-Validate mechanism (Source (Androulaki *et al.*, 2018))

The workflow of transaction in fabric is as follows. A high-level transaction flow is represented in Figure 3.5

- 1) Creation of the proposal: A client organization proposes or invokes a transaction request, and this proposal is sent to all the organization peers in the network as specified in the endorsement policy.
- 2) Transaction endorsement: Membership Service Provider (MSP) of the endorser peers where the proposal reaches will authenticate and authorize the identity of the transaction proposal followed by executing the chaincode (smart contract) to create a read-write set. The response is returned to the client. Figure 3.6 represents transaction endorsement design
- 3) Ordering Service: The orderers using RAFT protocol establishes ordering of transaction blocks and disseminate to all other organization peers.
- 4) Updating the Ledger: The peer nodes update their ledger with the received transaction block from the ordering service nodes. Thus, a new transaction gets committed.

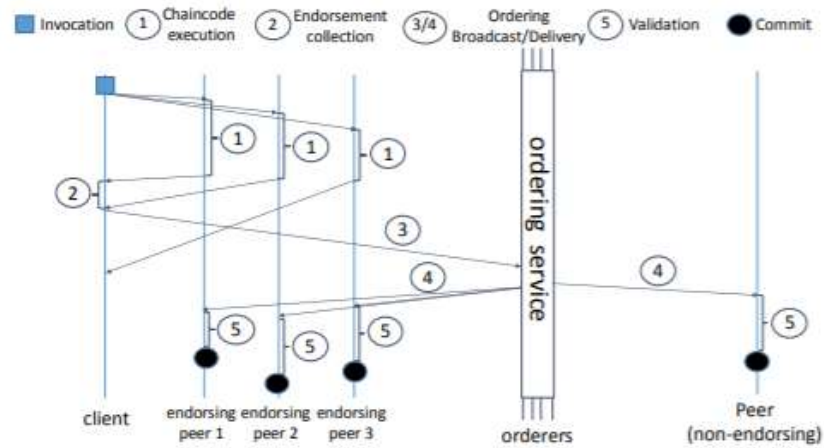


Figure 3.5. High-level Fabric transaction flow (Source (Androulaki *et al.*, 2018))

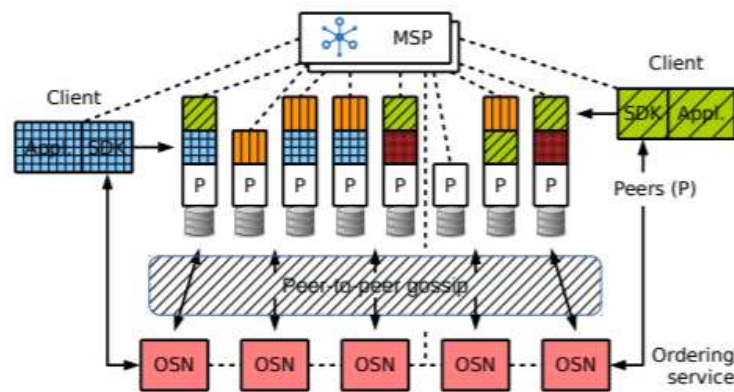


Figure 3.6. Transaction endorsement mechanism (Source (Androulaki *et al.*, 2018))

Private Data Collection (PDC)

Private data collection is a feature in Hyperledger Fabric where a subset of organizations in the same channel can protect their sensitive data from other organizations. The actual private data is stored in the PDC of the organization and the hash of that data is stored in the public state ledger. A three-phase transaction flow is shown in Figure 3.7. In the figure, peers P1 and P4 hold a PDC so that P1 and P4 can have only the private data, the rest of others contain only the hash of the private data (Wang *et al.*, 2021).

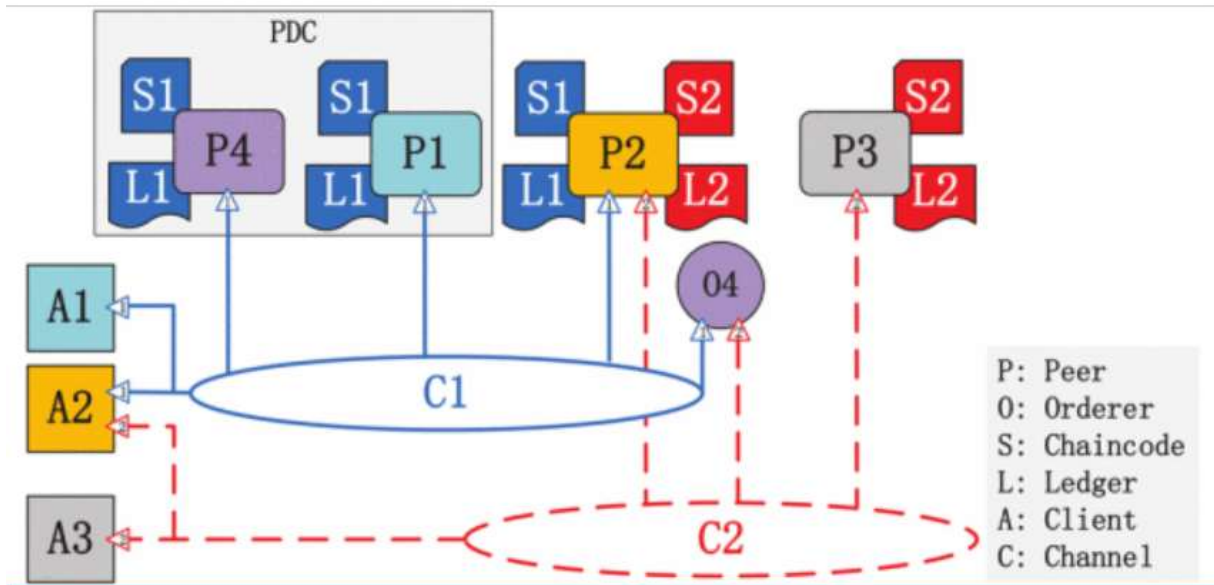


Figure 3.7. Three-Phase transaction flow (Source (Wang *et al.*, 2021))

AWS Nitro Enclave

AWS Nitro Enclave enables AWS EC2 instances to create isolated computed environments to protect and securely compute highly sensitive information such as healthcare data. Nitro enclave workflow is represented in Figure 3.9. Nitro Hypervisor technology is used in nitro enclave to create CPU and memory isolations. It is separated from the parent instance's kernel such that it doesn't have access to the enclave and a secure virtual socket (vsock) is used by the parent instance to interact with the enclave. AWS Key management service is integrated with the nitro enclave which enables encrypt-decrypt mechanism within the enclave. Nitro Hypervisor also handles the cryptographic attestation process to verify the identity of the enclave as well as the authenticity of the application running in the enclave (<https://aws.amazon.com/ec2/nitro/nitro-enclaves/features/>).

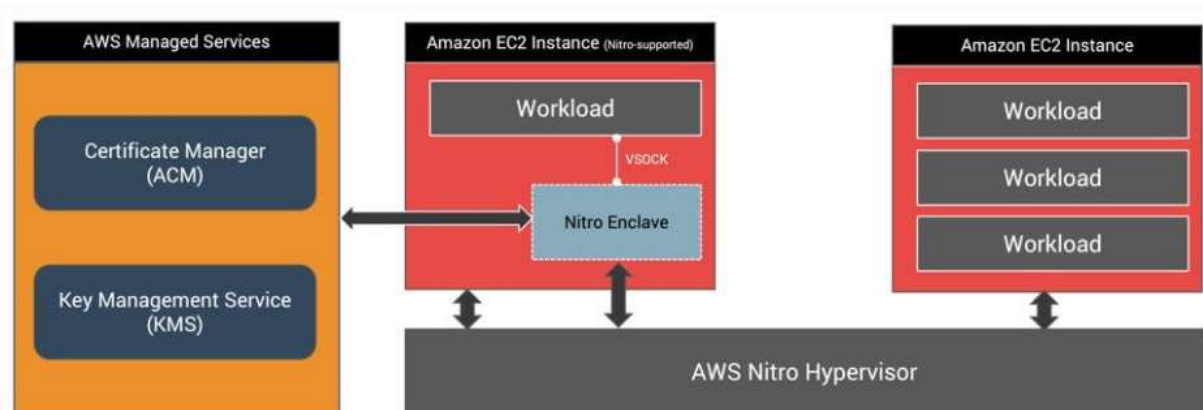


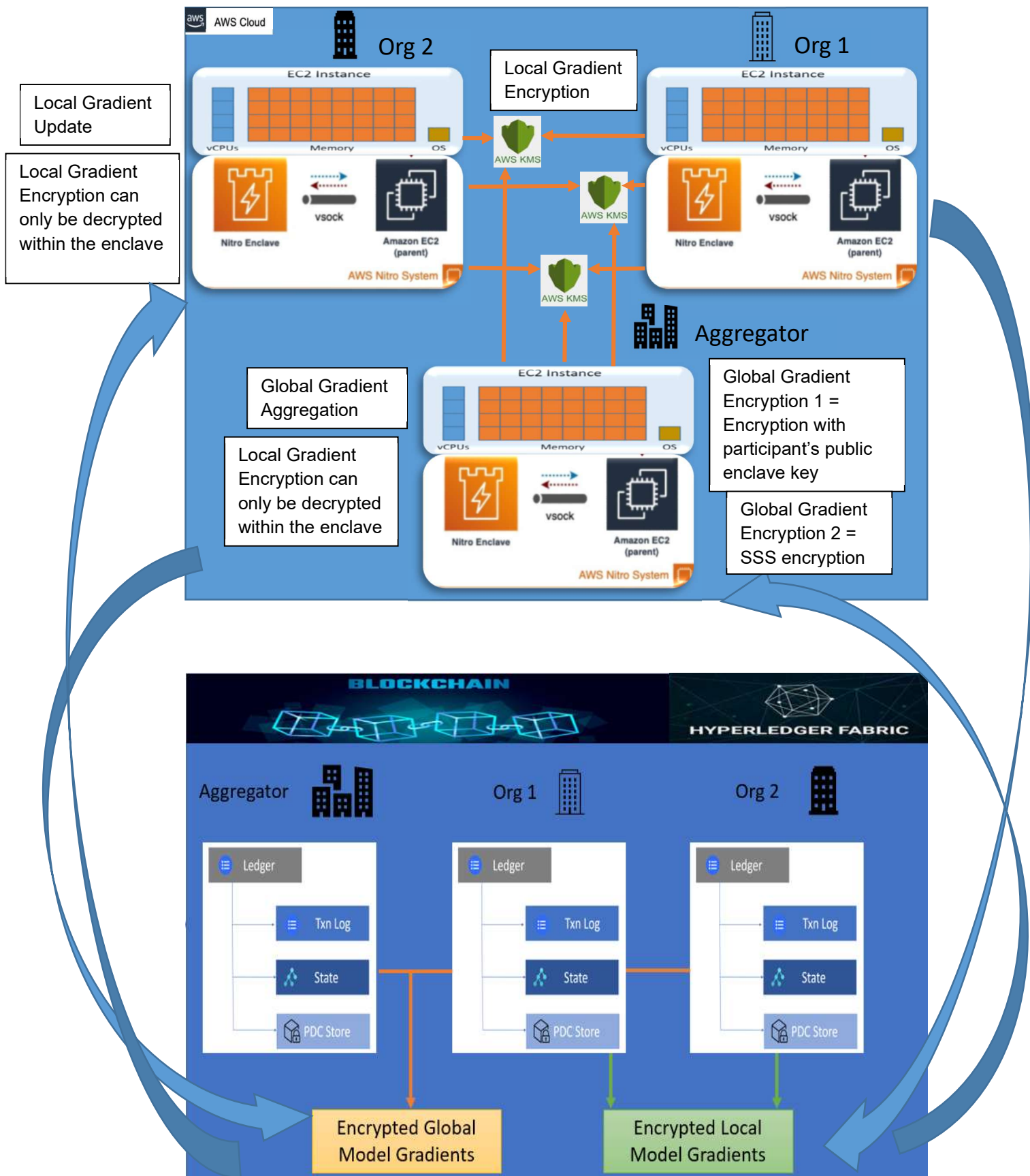
Figure 3.8 AWS Nitro Enclave

(Source <https://www.forbes.com/sites/janakirammsv/2020/10/30/aws-nitro-enclaves-bring-confidential-computing-to-amazon-ec2>)

The operation of the proposed Blockchain-FL framework consists of the following steps.

- 1) Aggregator's AWS parent instance starts the nitro enclave. The enclave creates public and private key for the enclave and send back the enclave public key to the parent instance.
- 2) The enclave public key is stored in the public ledger of Hyperledger.
- 3) The organization's AWS admin provides an access to aggregator to initiate a nitro enclave enabled instance within the organization's AWS account through a script automation process. The script is approved by all the organizations in the blockchain. This script enables to run the Local training with the organization's data while the global gradients are encrypted in the aggregator enclave within the organization account using enclave's private key.
- 4) The script starts with spinning up a nitro enabled EC2 instance and initializing the organization enclave where the enclave public key is sent to the PDC of Hyperledger blockchain.
- 5) The organization parent instance fetches the local model gradients and aggregator enclave public key from the blockchain and encrypts the gradients with the aggregator's enclave public key.
- 6) The encrypted local gradients are stored back in shared PDC of Hyperledger between the organization and aggregator.
- 7) In the aggregator instance, after all the organization's enclave public key and encrypted local model gradients are received, those are sent to the aggregator enclave.
- 8) The aggregator enclave decrypts all the local model gradients and compute the new global model gradients by taking the average of all the local gradients.
- 9) The updated global gradients are encrypted with each organization's public key and also with Shamir's Secret Sharing algorithm and send back to the parent instance of the aggregator.
- 10) The encrypted global model for each organization is stored in the respective PDCs and SSS encrypted global model is stored in the public ledger.

Orgs provide access to aggregator to initiate an instance for Local model training through a bash script which will be terminated after the process.



- 11) In each organization's enclave, the global model is decrypted with their enclave's private key.
- 12) In the parent instance of the organization, local model is updated with the global gradients.
- 13) The script terminates the process and the aggregator instance in the organization AWS account.
- 14) In case any of the client is not cooperating with the FL, the consortium will initiate a consensus process to remove the client and then initiate a smart contract to grant access for all the participants to retrieve the global model using SSS algorithm. The smart contract will get initiated only if the preceding value of number of participants in the network (before the termination of client) is greater than the current number of participants.

4.1 Summary

At this point of research stage, a blockchain driven FL framework has been proposed which opens a new path for a secure collaboration of healthcare corporations to necessitate in building better AI applications in medical field. Several analysis of system designs were considered in order to reduce communication overheads, computational costs and most importantly, security and privacy of sensitive healthcare data while performing federated learning. This paper demonstrates how a trusted execution environment can be easily automated and setup to launch an isolated computation environment with an encryption-decryption mechanism to defend poisoning attacks and information leakage attacks in FL. Furthermore, a consortium based blockchain combined with Shamir's Sharing Secret algorithm is leveraged to safeguard reprimands from 51% attack or Sybil attacks.

CHAPTER 4: IMPLEMENTATION

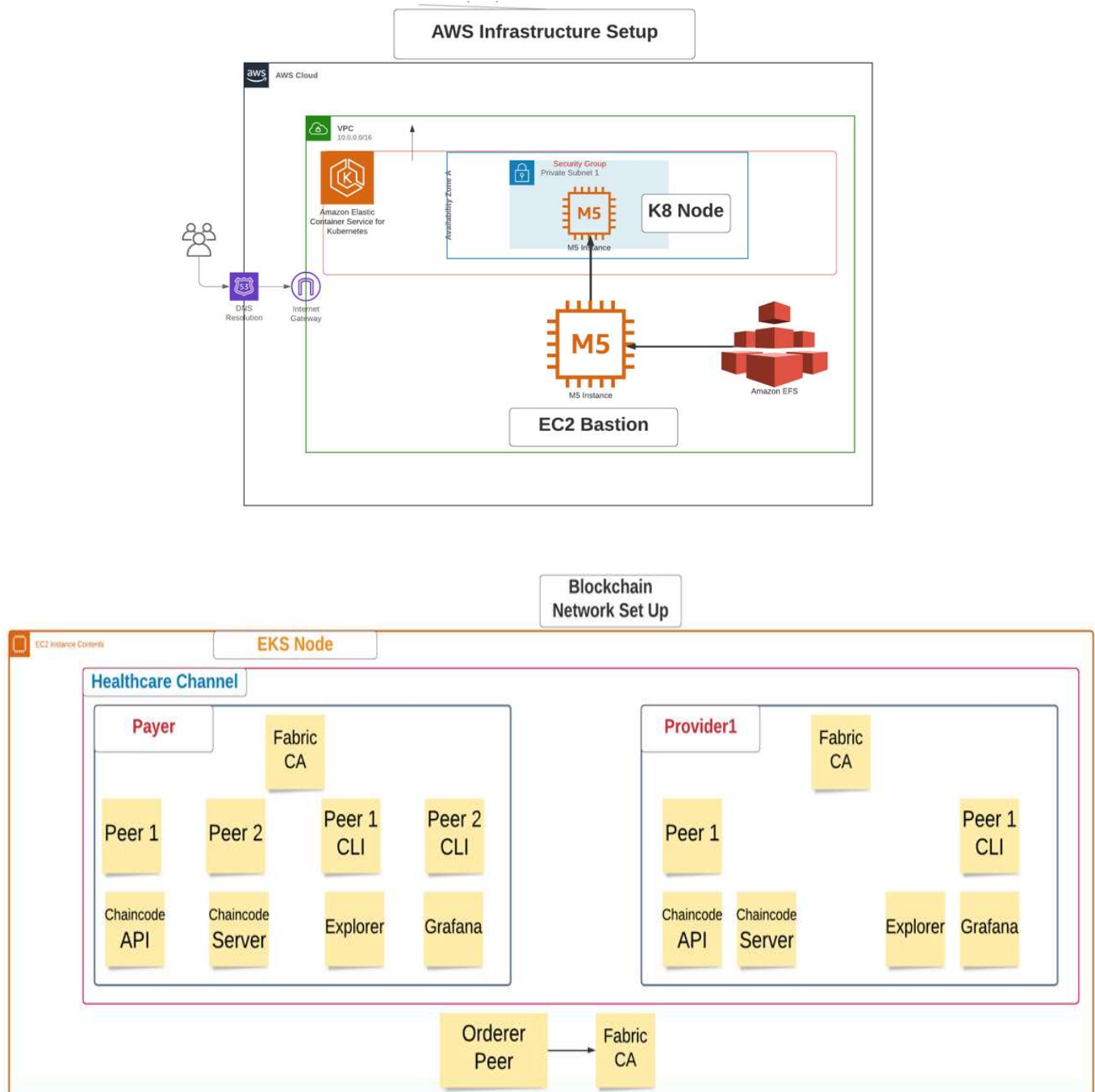
4.1 Introduction

This chapter discusses the architecture and implementation of the proposed Blockchain FL system. First, the assumption and setup are discussed and later a detailed technical architecture of the system is depicted. The technical architecture initially describes the implementation of the Hyperledger Fabric blockchain in a Kubernetes setting within the AWS cloud environment. Various types of resources used to setup the whole AWS infrastructure and blockchain network are also discussed. Then, this section outlines the structure and resources for federated learning environment. It also provides the cost estimation for the entire infrastructure setup including both hardware and software. Finally, it summarizes the implementation process.

4.2 Assumption and Setup

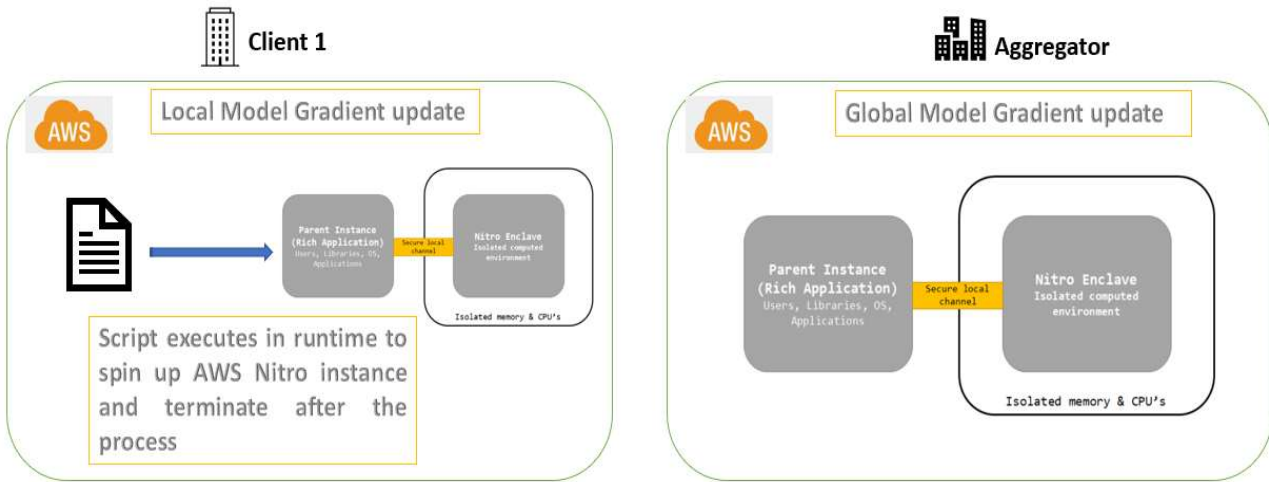
The experiments to be conducted in a distributed setting, a consortium blockchain network namely, Hyperledger Fabric was chosen where the network is assumed and set up in one same AWS region: North Virginia and 3 different virtual private clouds (VPC). Figure x shows AWS infrastructure setup. An EKS cluster was created for each of the clients, one aggregator (payer) and two clients (provider1 and provider2) to deploy Hyperledger Fabric in the Kubernetes setting for higher scalability as shown in Figure x (only provider1 is shown in the diagram). Fabric Certificate Authority (CA), Orderers, Peers, Peer CLIs, Chaincode and APIs are exposed as Kubernetes container pods in the EKS worker node for each organization. AWS Network load balancer was created to expose the orderer endpoint and anchor peers endpoints to different organizations to do the blockchain transactions. For the testing purpose, one EKS worker node, 2 vCPUs, 8 GB RAM m5x.large machine was created for each client to setup 2 peers each. A bastion, 2 vCPUs, 8 GB RAM m5x.large EC2 instance was created to access each private worker node.

AWS infrastructure is made up and running with the help of cloudformation and bash script. Blockchain network is setup using bash script and kubectl tool. A new client is added to the blockchain network by running a bash script. All blockchain operations are executed in the Peer CLI pod.



The Trusted Execution Environment (TEE) was realized by leveraging AWS EC2 instances, 4 vCPU, 16 GB RAM, m.xlarge with Nitro Enclave enabled for configuring isolated environments for sensitive data computing. Simple and straightforward setup of nitro enclave in AWS (as depicted in Figure x) facilitates unchallenging implementation of TEE environment in the cloud. In real time scenario, for the local training in each client, client-aggregator owned AWS nitro instances in the client's AWS account are spun in runtime with the help of an automation script (shown in Figure x), however, for testing purpose, this was done manually.

AWS Key Management Service (KMS) was used for TEE and other data (local and global models) encryptions. The global model learning is evaluated with the help of a two-layer convolutional neural network (CNN) with SGD as the optimization technique by utilizing the Tensorflow framework. The CNN softmax model uses a local batch size of 10, SGD momentum of 0.75 and learning rate of 0.01.



4.3 Experimental Settings

EKS Cluster 1 – Aggregator

Initially, the blockchain network is setup in the aggregator's cluster and is equipped to receive and store encrypted model gradients in the PDC and public ledger of Hyperledger. Figure x. shows the Kubernetes deployment for all the blockchain services in the aggregator. The codebase for the implementation can be found at <https://github.com/rijoblockchain/nitro-enclave-FL>. Succeeded by the blockchain network setup, AWS nitro instance for the aggregator is created

The parent instance of the aggregator waits to receive all the encrypted local gradients and enclave public key of each client as shown in Figure x. After receiving the same, the parent host sends those to the enclave through a secure Vsock channel. Within the enclave, all the local gradients are decrypted using aggregator's enclave private key. Then the average sum of gradients is computed and encrypt the newly updated gradients with each client's enclave public key. Also, the new gradients are encrypted with SSS algorithm. Hence, there are total 4 encrypted gradients. Three for each client (two organizations and one aggregator) and one with SSS encryption. Client's enclave public key encrypted gradients are stored in PDC and SSS encryption in state ledger of Hyperledger

EKS Cluster 2 – Client 1

Using a script, client 1 is added to the blockchain network set up by the aggregator. Client 1 provides necessary IAM permissions in their AWS account for the aggregator to spin up nitro enclave instances owned by both client and aggregator. The aggregator initiates the script for global model decryption within the enclave and local model training in the parent instance. The encrypted global gradient is fetched from the PDC of Hyperledger blockchain and local training data is taken from the S3 of the

client's AWS account. The updated local gradient is encrypted with aggregator's enclave public key and send to the state ledger. The nitro instance is terminated by the script after successful completion of local model training.

4.4 Implementation Cost

4.4.1 Hardware

Figure x. shows the hardware resource cost for the implementation of the proposed architecture.

Region	Service	Daily (USD)	Monthly (USD)	Configuration summary															
US East (N	Amazon EC2 Instance type (m5.large)	9.33	280.8	Operating system (Linux), Quantity (6), Pricing strategy (EC2 Instance Savings Plans 1 Year No Upfront), Storage amount (30 GB															
US East (N	Amazon EKS	2.43	73	Number of EKS Clusters (1)															
US East (N	Amazon Elastic File System (EFS)	0.08	2.4	Desired Storage Capacity (30 GB per month)															
US East (N	Amazon EC2 Instance type (m5.xlarge)	6.06	182.66	Operating system (Linux), Quantity (2), Pricing strategy (EC2 Instance Savings Plans 1 Year No Upfront), Storage amount (30 GB															
US East (N	S3 Standard	0.007	0.23	S3 Standard storage (10 GB per month)															
US East (N	AWS Key Management Service	0.366	11	Number of customer managed Customer Master Keys (CMK) (5), Number of symmetric requests (2000000)															
Total		18.27	550.09																

4.4.2 Software

Software	Type	Cost
Visual Studio Code	Open source	Free
Lens IDE	Open source	Free
JupyterLab	Open source	Free
Pycharm	Open source	Free
Tensorflow framework	Open source	Free
Docker	Open source	Free
Golang	Open source	Free
Nodejs	Open source	Free

4.5 Summary

This chapter provided a description of the technical side of the research project, including its implementation side and design decisions. The development phase was a mixture of top down and bottom up. The general outline and required structure of the architecture were developed in the design

phase. The actual development process was bottom up due to the fact that the technology was new and the learning curve was sharp. As the research advanced, once the possibilities of the technology had been comprehended, refinements and enhancements were made, but no underlying design elements were altered. In this chapter, setup of infrastructure and blockchain network was discussed along with the implementation cost for the infrastructure resources. In addition, the experimental setting was also referenced.

CHAPTER 5: TESTING AND EVALUATION

5.1 Introduction

The goal of this segment is to reviews the test results and evaluates the security and privacy of AI models especially the model gradients in the healthcare federated learning settings. Based on the threat model designed, the evaluation will be conducted on two types of attack scenarios – Untargeted & Targeted attacks and Privacy leakage attacks (adversarial attacks). It explains how the framework could withstand these attacks without using any auxiliary privacy-preserving techniques leveraging TEE and blockchain technology. This section also demonstrates a malicious activity by an attacker and how the system handles it effectively. Finally, accuracy comparison of proposed method and differential privacy is put on view.

5.2 Attacks on Proposed Framework

5.2.1 Attack Type Scenario 1 - Untargeted and Targeted Attacks

Targeted attacks aim to affect the minority of data items in the model while maintaining the performance of the model (Bagdasaryan *et al.*, 2020b) whereas untargeted attacks inject false value or incorrect subset of data in the local dataset to alter the learning outcome of local clients (Li *et al.*, 2019; Peng, Li and Ling, 2021). Examples of targeted and untargeted attacks are backdoor attacks & gradient-ascent attacks, and sign-flipping & same-value attack respectively.

The run time automation of spinning up the TEE enclave in AWS and the encryption-decryption mechanism within the enclave along with the storing of encrypted model gradients in the blockchain makes it difficult for the attackers to instil poison attacks in the local model. Also, the immutable and consensus features of blockchain alleviates unnecessary malicious changes in the local model encrypted gradients due to stringent blockchain administrative process. The results showed in the figure x demonstrates that an attacker intends to manipulate the blockchain's couchdb database by changing the value of the encrypted model gradients from the organization's account. However, the attacker failed to accomplish the operation due to the inherent secured nature of blockchain.

The overall classification performance of a traditional differential privacy method under poison attack is also evaluated. Figure x shows the experimental results. The results depicts that the method used in this research should outperform all the methods using differential privacy irrespective of the proportion of malicious attackers. The differential privacy method fails to converge and obtains low accuracy as the proportion of poison attackers increases. Poisoned attacks highly amplify the model parameters that leads the global model to deviate abruptly. Using the framework proposed in this research, the result is superior to federated learning with no poison as the possibilities of instilling poison attacks are very less due to the careful design of the framework architecture leveraging blockchain and TEE, thus mitigating

communication overheads, computational costs and curtailed model accuracy due to additional privacy-preserving mechanisms.

5.2.2 Attack Type Scenario 2 – Privacy Leakage

The aggregator and the clients' AWS Nitro Enclaves are fully isolated virtual machines, hardened, and highly constrained. It doesn't have persistent storage to store any model gradients, no interactive access, and no external networking so that it makes root user or an admin user on the parent instance to access or SSH into the enclave. Transferring of encrypted gradients and enclave public keys between parent instance and the enclave is done using a secure local channel called Vsock. Moreover, Nitro Enclaves uses the proven isolation of the Nitro Hypervisor to further isolate the CPU and memory of the enclave from users, applications, and libraries on the parent instance. These features help isolate the enclave, and significantly reduce the attack surface area.

Cryptographic Attestation allows the aggregator and the clients to verify the enclave's identity and that only authorized code is running in the respective enclaves. The attestation process is accomplished through the Nitro Hypervisor, which produces a signed attestation document for the enclave to prove its identity to the aggregator or the participants. Attestation documents contain key details of the enclave such as the enclave's public key, hashes of the enclave image and applications, and more. Therefore, the model gradients encrypted with the enclave's public key can only be decrypted within the enclave with its own private key which makes it more secure and avoids any adversarial attacks on the private information of the patients.

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

5.2 Introduction

6.2 Discussions and Conclusions

This research paper addressed data privacy issues in federated learning process by integrating blockchain with Trusted Execution Environment (TEE) to federated learning framework. This research endeavours to not use traditional privacy preserving techniques such as differential privacy, homomorphic encryption and multi-party computation to avoid communication overheads, inefficiency and accuracy concerns. A TEE environment is setup leveraging Amazon Web Service cloud platform's Nitro Enclave enabled EC2 instance where the encryption and decryption process of global and local model gradients is done within the enclave using its own public and private keys and no users even the admin of the parent instance doesn't have access to the enclave. In addition, a consortium based blockchain architecture is employed leveraging Hyperledger Fabric to ensure the auditability and immutability of the encrypted local and global models. Hyperledger Fabric network is setup in a Kubernetes setting for higher scalability and adding of a new organization in the network is achieved with an automated script. Shamir's Secret Sharing algorithm is also used to encrypt the global model gradients within the enclave to ensure smooth approval process in the blockchain in case any organization is reluctant to participate in the federated learning process. The results of analyses and experiments demonstrates that the proposed blockchain based federated learning framework achieves

good accuracy, high robustness and enhances privacy and security. The test result also shows the accuracy of the model compared with models with applied differential privacy methods.

6.3 Key Contributions/Contribution to knowledge

Aws nitro enclave – TEE easy to spin up with no additional cost in a cloud platform. No separate hardware required.

Easy for practical implementation

Can reduce communication overheads, costs and encryption inefficiencies and accuracy

Scalable blockchain network in Hyperledger fabric using Kubernetes with automated script for setting up AWS infrastructure and the blockchain network and adding new organization in AWS infrastructure

6.4 Future Recommendations

For local model updates, an automation script is used to spin up the infrasturucte in run time. However, any back channel attack to leak the model gradients are not considered in this research and is a work for the future.

As of now, AWS nitro enclave doesn't support GPU virtual machines which restricts tensorflow to run inside the enclave machine. That's the reason, only gradient encryptin is done within the enclave. As per the information from AWS, they venture to add GPU functionalities in the enclave so that the whole federaed learning process can be executed inside the enclave. This can avoid the script part in this research. Hence, the future work will be based on executing FL within the enclave.

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APPENDIX A: RESEARCH PROPOSAL

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Research Proposal

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Abstract

Artificial Intelligence (AI) and Blockchain plays a tremendous role in the healthcare domain, in which AI exceptionally aids doctors to detect, predict and diagnose diseases with excellent accuracy and cost-efficiently. AI dispenses knowledge databases by storing and processing enormous data that facilitates superior disease prediction and thus enhancing clinical decision support. Even so, AI systems are not invulnerable. These systems require training of massive data to achieve better outcomes and lack of sufficient data might lead to biased prediction errors

which could result in health-related consequences. Data is fragmented in silos across multiple sources like electronic health records (EHR) in healthcare providers, insurance companies, pharmacies etc. by which data gets complex and less comprehensible to AI systems. An efficient way to collaborate with these organizations and to access and share data without compromising security and confidentiality of patients is the need of the hour. To overcome the challenges of sharing sensitive patient information for AI development, federated machine learning (FL) can be leveraged to an extent. Federated learning is a distributed and decentralized machine learning technique to collaboratively train the model without exposing the real data, instead they share model gradient parameters. However, studies emphasize that conventional FL opens gaps in terms of privacy leakage that needs immediate attention. Blockchain technology is a propitious solution to the aforementioned security and privacy concerns due to its immutable, decentralized and trustless behavior. A private consortium based blockchain framework can be utilized to develop a consensus driven decentralized EHR systems to hold personal and highly sensitive health data, thus mitigating data access and sharing plights. This research focuses on combining Blockchain and Federated learning to address the underlying privacy issues.

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LIST OF ABBREVIATIONS

AI.....	Artificial Intelligence
BC.....	Blockchain
EHR.....	Electronic Health Record
FL.....	Federated Learning
PBFT.....	Practical Byzantine Fault Tolerance
PoW.....	Proof of Work
PoS.....	Proof of Stake
HIPAA.....	Health Insurance Privacy and Protection Act
GDPR.....	The European General Data Protection Regulation
HL7.....	Health Level Seven
FHIR.....	Fast Health Interoperability Resources
IPFS.....	InterPlanetary File System
DICOM.....	Digital Imaging and Communications in Medicine
BFL / BCFL...	Blockchain based Federated Learning
PDC.....	Private Data Collection
AWS.....	Amazon Web Service

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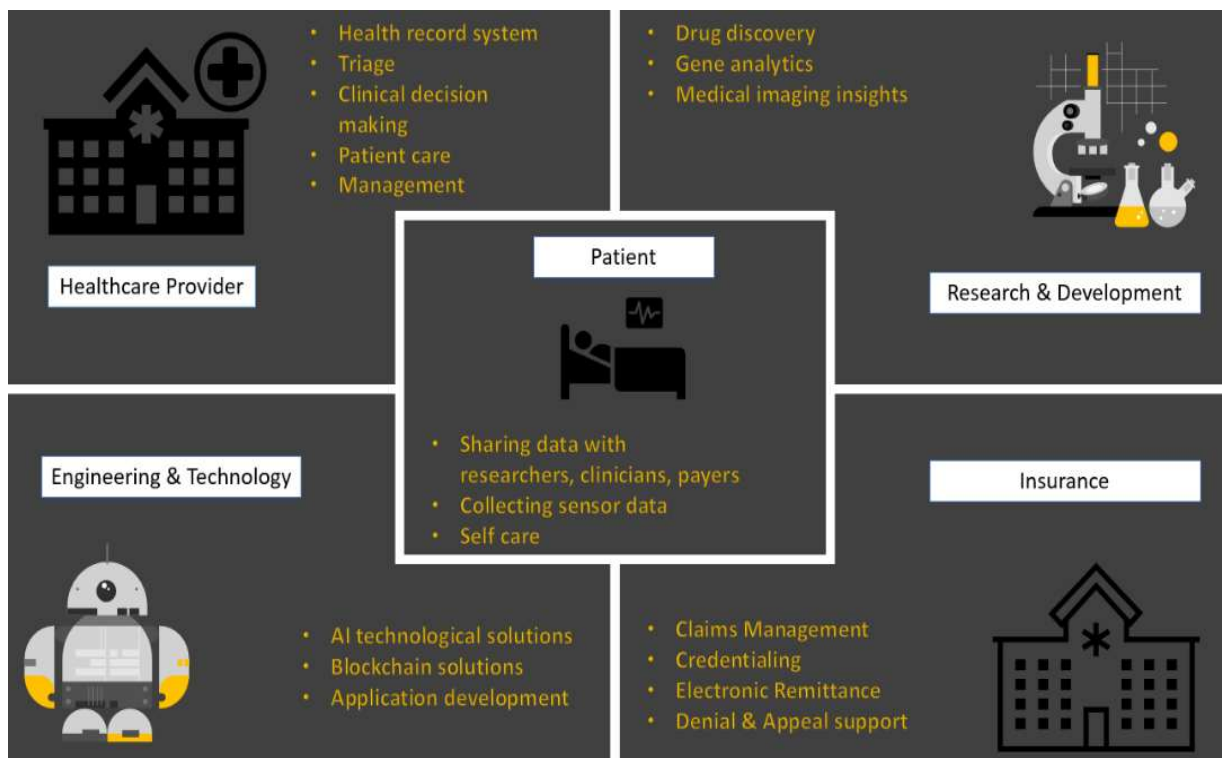
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1. Background

Blockchain technology is shaping a new revolution in the healthcare eco system with its cryptographic, transparent, immutable, distributed, and trustless nature. By fully utilizing this cutting-edge technology, this decade will witness the evolution of healthcare industry to a patient-centric model with enhanced security, privacy, and interoperability of health data. One of the prominent use cases of blockchain in the healthcare is medical record access and sharing. Blockchain powered healthcare applications can employ patient-controlled accessibility to put together patient records easily, thereby enabling hassle-free exchange of data across different institutions (Hölbl *et al.*, 2018).

The healthcare domain is a sensitive-data intensive sector and always challenge-driven. Access control, data integrity, provenance and interoperability are critical to maintain both patient's privacy and data exchange between disparate systems (Hasselgren *et al.*, 2020). To achieve desired health results, collaboration between multidisciplinary health teams is imperative, as shown in Fig 1. Therefore, a secure medical data exchange is crucial in this regard. Considering all these aspects to store and share sensitive medical data, blockchain is a promising framework for healthcare data.

Figure 1.1. Collaborations in healthcare



There exist three types of blockchain namely public, private and consortium or federated (Morkunas, Paschen and Boon, 2019). Table 1.1. shows the summary of types of blockchain. A consortium based private blockchain is preferable for healthcare use cases as it offers additional degree of transaction privacy, which is critical for sensitive data transactions. Moreover, consensus mechanisms like PBFT facilitate greater transaction throughput, scalability and cost cutting. Since only verified parties are allowed to initiate future participants in the consortium, it adds higher level of trust, reliability and security. Table 1.2. shows an overview of consensus mechanisms.

Table 1.1 Types of blockchain summary (Hasselgren *et al.*, 2020)

Property	Public Blockchain	Consortium Blockchain	Private Blockchain
Consensus Nodes	Miners	Selected nodes	Governance Structure
Transparency	Public	Public or restricted	Public or restricted
Immutability	Nearly impossible	Partial	Partial
Throughput	Low	High	High
Energy Efficiency	Low	High	High
Centralized	No	Partial	Yes
Authentication	Permissionless	Permissioned	Permissioned

Table 1.2 Types of consensus algorithms overview (Xiao *et al.*, 2020)

Property	Proof of Work (PoW)	Proof of Stake (PoS)	PBFT
Structure	Permissionless	Permissionless	Permissioned
Energy consumption	High	Medium	Low
Adversarial hashing power toleration	< 25% computing	< 51% stake	< 33.3% faulty replicas
Transactions per sec	7 - 30	30 - 175	Upto 20000
Example	Bitcoin	Dash	Hyperledger Fabric

Artificial Intelligence (AI) is a disruptive technology that has the potential to transform healthcare industry and is poised to be the driving factor that aids to gain unprecedented insights about disease diagnostics and patient care spectrum. The application of AI in medical imaging has opened a new era to facilitate detection of even small abnormalities and other extensive clinical conditions with impressive accuracy, leveraging AI techniques such as computer vision, traditional machine learning and deep learning (Kaul, Enslin and Gross, 2020).

For the medical AI applications to offer valid, accurate and effective results, larger amount of imaging and clinical data is required. Even though, the world's patient databases contain enough data to acquire promising outcomes, the inadequacy of systematized electronic medical records across different medical facilities poses a significant barrier to data availability for constructive algorithm training. Moreover, stringent laws and regulations such as Health

Insurance Privacy and Protection Act (HIPAA) and The European General Data Protection Regulation (GDPR) to protect patient privacy still poses a security concern to access these data which causes a significant hindrance to AI development (Kaissis *et al.*, 2020) (Liddell, Simon and Lucassen, 2021). As the advent of Health Level Seven (HL7) Fast Health Interoperability Resources (FHIR) Release 4, a new standard for better interoperability to exchange health data can be secured (Ayaz *et al.*, 2021a). Even though with the implementation of FHIR standards, the current medical systems destitute transparency, provenance, trust, security, and privacy (Yaqoob *et al.*, 2021).

A secure and privacy-preserving AI approach can be achieved by employing decentralized data and federated machine learning (FL). FL is an encrypted privacy-preserving decentralized collaborative machine learning technique in which the data owners collaboratively train a model and in this process none of the data owners expose its real data, instead share only the training gradients. This technique enables the training of algorithms on data while the data remain with its owner (Yang, Liu, Cheng, *et al.*, 2019).

Nevertheless, traditional federated learning comes with security and privacy risks such as single-point failure, poison attack, privacy leaks and lack of motivation. Even though, various methods have been devised to eliminate these issues, it falls short of communication latency and computational performance. While Blockchain driven federated learning has shown great plausibility in addressing these threats, inferring attacks (Z. Wang *et al.*, 2019a) by applying back deduction is still possible even in blockchain to reconstruct the patient information which is a clear case of privacy breach.

This research tenders a study on a novel architecture for federated learning within a permissioned private healthcare consortium based on Hyperledger Fabric framework. The idea is to contain all healthcare assets including patient clinical records, medical images, de-identified data, local training model gradients and global model gradients within the blockchain. Since the private consortium comprises only of trusted parties and all transactions are logged in the blockchain, a malicious act from any parties could be alleviated.

2. Problem Statement

2.1 Scalable EHR with implementation of FHIR in a blockchain system

The HL7 FHIR (Fast Healthcare Interoperability Resources) standard allows electronic health records, including clinical and administrative data, to be exchanged between various data silos. (Antwi *et al.*, 2021a) proposed an EHR solution in a permissioned blockchain that satisfies criteria for GDPR and HIPAA and implemented the same leveraging Hyperledger Fabric Composer, while (Asad *et al.*, 2020) proposed a framework for blockchain based healthcare data sharing with permission-based access that adheres to FHIR standards and tested it on Multichain blockchain. The study in (Lee *et al.*, 2020) suggested a blockchain architecture that ensures the security, privacy, integrity, and availability of the data incorporated with FHIR's data format standards. (Lee *et al.*, 2020) presents a privacy-preserving EHR management solution that uses Hyperledger Fabric permissioned blockchain framework to store patient records effectively whilst providing anonymity and unlink-ability. They also evaluated the performance metrics of data transaction on-chain. (Bittins *et al.*, 2021) proposed an integration of FHIR server with Hyperledger fabric blockchain to store EHR data where EHR data resource is created, updated and deleted in FHIR server database and the corresponding transaction log is stored in fabric.

2.2 A Blockchain solution to prepare and share Medical Images for AI-driven image analysis

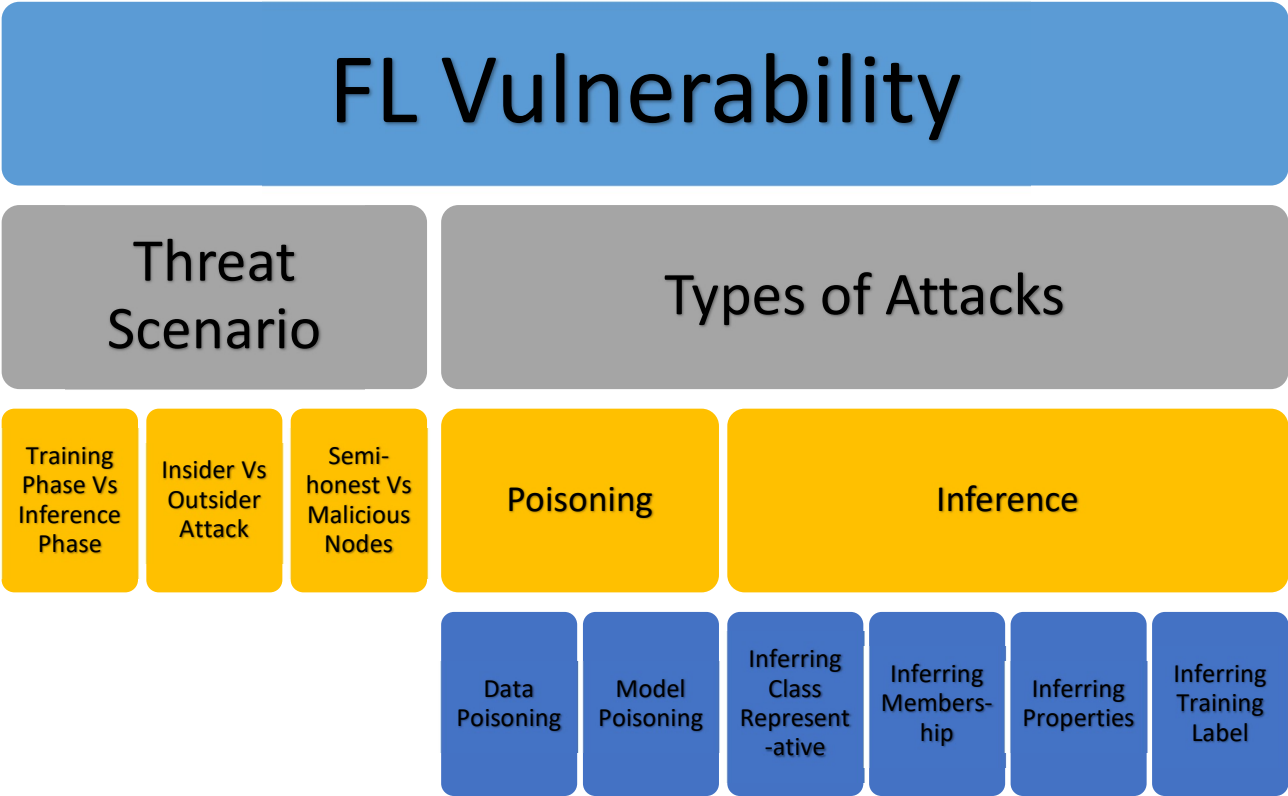
There are certain steps needed to be taken before medical images can be used for the development of AI algorithm, such as ethical approval, data access to unprocessed image data source, querying data, data de-identification, data transfer, etc. as specified by (Willemink *et al.*, 2020). The research paper (Sridevi *et al.*, 2021) depicts a proposed architecture to integrate cloud and blockchain to store medical images. Here, blockchain is used to store only the hash of images and the original images are put into cloud. (Patel, 2019) provides an overview of a blockchain implementation to enable a patient-controlled and decentralized sharing of medical images. (Rajesh Kumar *et al.*, 2021) leveraged InterPlanetary File System (IPFS) distributed storage platform to store medical images for analysis and hash of the images are stored in the Ethereum network. (Kumar and Tripathi, 2021) shows how blockchain can be utilized to share DICOM images in a healthcare application while maintaining immutability, confidentiality, and transparency of information. The researchers in the paper (Tan *et al.*, 2021) using Hyperledger

fabric demonstrated a blockchain-enabled AI platform to securely share datasets and deep learning algorithms and to report deep learning algorithm validations and test results.

2.3 Blockchain based federated machine learning

The federated learning approaches combined with blockchain technology have arguably turned out to be the forthcoming privacy preservation technique in medical AI applications. In blockchain based FL, blockchain is used to store only the local and global model gradients and parameters, but not the entire data (R Kumar *et al.*, 2021). (Aich *et al.*, 2021) proposed a BCFL solution to protect personal healthcare record where blockchain is used to store and verify the accessibility of data, which is stored in an external database, while (Passerat-Palmbach *et al.*, 2019) introduced an architecture for federated learning in healthcare consortia in Ethereum blockchain. However, federated learning is not invincible to security and privacy threats. Back deduction while the transmission of gradient parameters has an adverse effect on patient’s confidentiality. Thus, FL is not adequately secure. A detailed overview of vulnerabilities is shown in Fig 2.3.1. The general attack types are categorized into Data poisoning attack, Model poisoning attack and Inferring attack (L. Li *et al.*, 2020a) (Zhang *et al.*, 2020).

Figure 2.3.1. An overview of FL Vulnerabilities. Source (Jatain, Singh and Dahiya, 2021)



Although numerous privacy-preserving approaches for traditional federated learning have been proposed and implemented as shown in this paper (Yang, Liu, Chen, *et al.*, 2019), further measures are necessary to expand its privacy-preserving scope in terms of single-point-failure, communication overheads, efficiency-security trade-off, computational performance, and lack of incentives. (Z. Wang *et al.*, 2019a) (Wang and Hu, 2021).

Table 2.3.1. shows summary of different BFL architectures. (Shayan *et al.*, 2021a) has brought forward a solution to provide privacy-preserving P2P FL through a distributed blockchain ledger, eliminating single point failure, poison attack and privacy leakage. FLChain (Bao *et al.*, 2019a) is a BFL platform with trust and a reward mechanism to motivate the trainer to be honest and to punish the dishonest node. An incentive is accorded to the nodes based on the size and quality of data samples that affects the accuracy of FL (Kim, Park, Bennis and S. Kim, 2020). DeepChain (Weng *et al.*, 2019a) achieved three security goals – confidentiality for privacy of local gradients, auditability and fairness. (Kang *et al.*, 2019a) stated that a contract-theory based incentive scheme in a consortium blockchain can entice high-reputation clients that have high accuracy with high-quality local training data to ensure reliable FL, thereby removing compromised and malicious workers.

Table 2.3.1. Different types of BFL architectures with solved issues. Source (Hou *et al.*, 2021b)

Architecture	Solved issues	Type of blockchain	Consensus mechanism
BlockFL [7]	Single point failure, lack of motivation	Public	PoW
FLChain [8]	Single point failure, lack of motivation	Public	–
FLChain [9]	Single point failure	Public	PBFT/PoW
RFL [10]	Poison attack, lack of motivation	Consortium	PBFT
DeepChain [11]	Single point failure, lack of motivation, privacy leakage	Public	Blockwise-BA
FedBC [12]	Single point failure, privacy leakage	–	–
BC-FL [13]	Single point failure, lack of motivation	Public	Pow
BlockFLA [14]	Single point failure, poison attack, lack of motivation	Public & private	PoW & PBFT
Chain FL [15]	Single point failure	Private	PoA
PSFL [16]	Poison attack	–	–
BFEL [17]	Single point failure, poison attack	Public & consortium	DPoS/PBFT & PoV
Biscotti [18]	Single point failure, poison attack, privacy leakage	public	PoF
VFChain [19]	Single point failure, poison attack	Public	Algorand
BLADE-FL [20]	Single point failure	Public	PoW
BFLC [21]	Single point failure, poison attack, lack of motivation	Public	Committee
VBFL [22]	Single point failure, poison attack, lack of motivation	Public	PoS

This paper initially attempts to build a permissioned consortium blockchain based EHR system that is highly scalable in terms of performance and storage to reserve sensitive patient data including reports, financial documents, lab results, vital signs, medical images, de-identified images, local and global model gradients on-chain. The architecture will utilize the private data collection (PDC) and key-level endorsement policy features of Hyperledger fabric to implement

a GDPR and HIPAA compliant FHIR resources to protect sensitive patient information and achieve patient-controlled access. On the contrary to the related work (Kang *et al.*, 2019a) that claims that reproducing FHIR resources and storing asset in blockchain is complex and useless, (Tanwar, Parekh and Evans, 2020) concluded that storing assets in blockchain offers the potential to improve the verification and integrity of rapidly expanding healthcare data. The performance and storage scalability (J. Xie *et al.*, 2019) of these enormous healthcare records especially images along with FHIR release 4 standards in blockchain still manifest a research space. One of the aims of this study intends to fill the aforementioned gap.

After the healthcare consortium is formed, the next stage is to relate federated learning with the data in the blockchain. Numerous blockchain based federated learning techniques have been proposed to overcome existing issues in tradition FL such as single-point failure, poison attack, lack of motivation and privacy leakage, for example (Shayan *et al.*, 2021b). However, as mentioned in the SLA (Hou *et al.*, 2021a) to prevent privacy leakages, BFL architectures must employ additional privacy-preserving methods that lead to communication overheads, efficiency-security trade-off and computational performance. Hence, within the blockchain technology, there exists a gap between confidential data protection and data availability for AI developments and bridging this gap is another area of interest in this research.

3. Research Questions

The following research questions are suggested for each of the research objective as highlighted as follows.

3. What can be done to reinforce the security and privacy of sensitive data in the healthcare domain while complementing enhanced interoperability especially for AI development?
4. Could a consortium based blockchain EHR system resolve the issues stated in the above question with improved performance and storage competence?
5. How can a consortium blockchain mitigate the privacy risks in a distributed Federated learning environment?

4. Aim and Objectives

The main aim of this research is to propose a permissioned blockchain driven EHR system that realizes performance and storage scalability to store sensitive patient data including imaging studies within a healthcare consortium. The healthcare assets in the blockchain will adhere to FHIR 4.0 standards and privacy regulations such as GDPR and HIPAA. These medical data will be further used by the organizations in the healthcare consortium to apply federated learning for data and image analysis, thereby reducing security and privacy risks.

The research objectives are formulated based on the aim of this study, which are as follows

- To study and implement a healthcare consortium blockchain framework to store patient's clinical records and medical images (EHR) on-chain that satisfies privacy laws and regulations with the realization of performance and storage scalability
- To implement HL7 FHIR 4.0 interoperability standards to reserve data within the consortium which facilitates hassle-free data exchange outside the blockchain system.
- To define the framework for blockchain based FL where the clients of FL are the nodes of blockchain who are the participants in the local model training as well as the global model aggregation.
- To develop a reward system to incentivize the participants basis their contribution to the model optimization and to compensate the patients who provide consent to share their data.

5. Significance of the Study

This research will be a significant endeavor in addressing a wide array of challenges in healthcare industry such as care coordination, data security, privacy, and interoperability issues. Many researches observed to log only the transaction in blockchain and the data to be kept in an external databases which are vulnerable to cyber-attacks. Through this research, the governments and healthcare communities will realize the benefits of a permissioned on-chain solution to store data, share medical data, protect sensitive information, and provide patients control over their data while keeping the data compliant to existing privacy regulations (GDPR and HIPAA). As the health regulatory bodies are tightening the mandates for the use of FHIR standards in data exchange between different systems, an attempt to implement FHIR resources

in the blockchain ensure smooth interoperability within and outside blockchain networks. Addressing the technical challenges like performance and storage scalability to store massive data in the blockchain will entice the medical and blockchain community to promote the use of blockchain in healthcare industry.

This study will provide new insights into decentralization of federated learning process to overcome the risks inherent even in blockchain driven FL such as privacy infringements. Mitigating these risks combined with an effective incentive scheme for local model training will attract more organizations, research institutions and healthcare providers to form a healthcare consortium. As the consortium expands, data become large. More data leads to efficient AI developments.

Moreover, the studies, analyses and demonstrations presented in this research will push for new paradigms which will be useful for further studies to innovate current strategies being employed in healthcare blockchain.

6. Scope of the Study

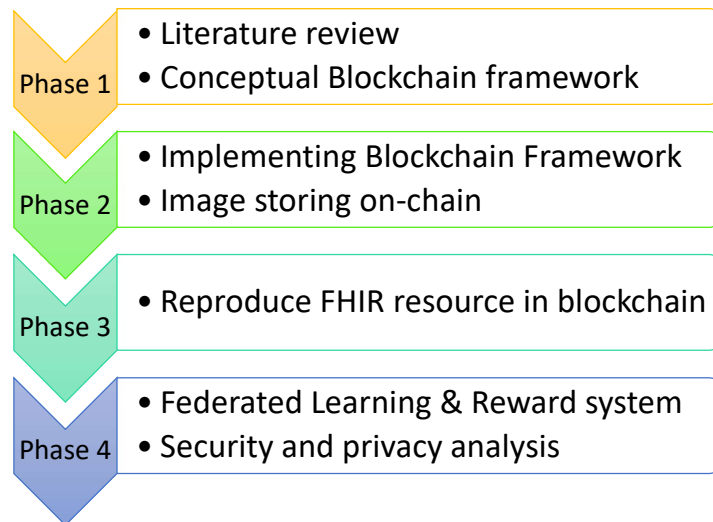
The study primarily focuses on developing a healthcare consortium blockchain to store healthcare assets and to securely share data for federated learning. For the study purpose and simplicity, the consortium will be confined to maximum 3 organizations. Technically, the research will be carried out in only one AWS account with all the organizations set up in the same cluster. As the study aims to investigate the security and privacy preservation of data in federated learning, accuracy improvement of FL results will not be considered. Hence, this research will attempt to utilize an existing FL method and an incentive mechanism. Reproducing all the FHIR resources in the blockchain is an extremely time-consuming process. Therefore, a few major FHIR resources, for the sake of this research will be implemented.

7. Research Methodology

This research proposes to develop a methodology, through theoretical research as well as a PoC implementation of the proposed architecture. The project will be advanced in 4 phases to reach the objectives as indicated in Fig 2: (1) Developing a conceptual blockchain framework; (2)

Implementing a scalable blockchain framework for healthcare assets; (3) FHIR resource implementation; (4) Applying federated learning with a reward mechanism and security analysis of private data.

Figure 7.1. Research Phases



In the first phase of the research, four main parts of a literature review are conducted, after which a conceptual blockchain framework for storing medical assets is designed. Firstly, to understand deeply the challenges in developing an on-chain private data repository in a healthcare consortium especially with the privacy laws and regulations in place (McGhin *et al.*, 2019) and examine the improvisation aspects in the existing methods, for example (Antwi *et al.*, 2021b). Secondly, to confront the issues with respect to the performance and storage scalability to store humongous health data in blockchain and how scalability could be achieved for excessive read and write operations (Stamatellis *et al.*, 2020). Thirdly, to comprehend on reproducing FHIR resources in blockchain (Ayaz *et al.*, 2021b). Fourthly, challenges in security and privacy in blockchain based federated learning will be studied (Hou *et al.*, 2021b). Finally, the conceptual permissioned consortium blockchain framework for storing assets and federated learning will be developed.

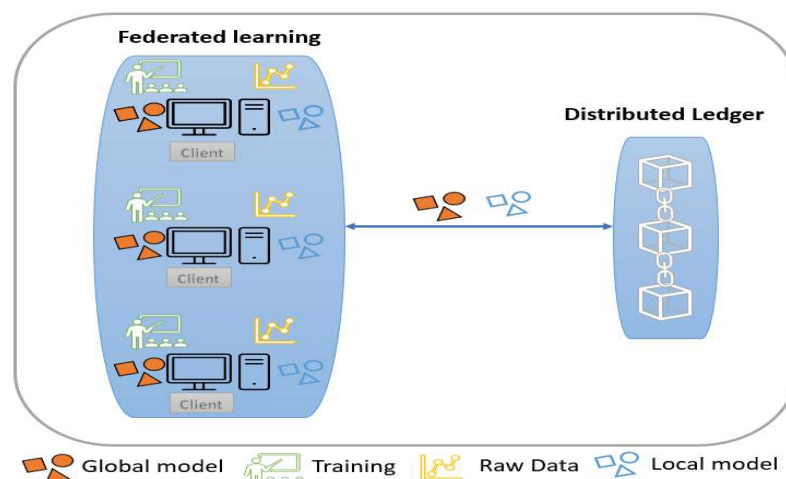
The second phase of this research will involve the implementation of healthcare consortium framework in a cloud platform namely AWS. To form the blockchain network specifically Hyperledger, an attempt will be observed to utilize some services in AWS such as EC2, VPC and S3. This phase will also explore the performance and storage scalability of large data by leveraging services like Elastic Kubernetes Service (EKS) and Elastic File System (EFS) respectively in AWS cloud. The BFT consensus mechanism, Fabric Certificate Authority,

private data collections (PDC) and key-level endorsement policy features of Hyperledger fabric will be investigated to implement a patient controlled electronic health record system. This research phase will also investigate the methods for storing medical images on-chain, say, in base64 format (Antwi *et al.*, 2021b).

The third phase will employ FHIR Release 4 standards to the healthcare data in the blockchain system as conforming to certain privacy and data standard regulations. This will ensure easy data exchange between disparate systems, in this case, within and exterior to blockchain.

The last phase of this study mainly focuses on devising a federated learning approach and to implement an incentive mechanism for the participants in the consortium. A fully decentralized FL model such as FuC-BCFL (Wang and Hu, 2021) is envisaged to be implemented where every organizations in the consortium will have the chance to participate in the local training model and global model aggregation. Fig 3, depicts a preferable FL architecture where the clients or workers using their data, trains the model locally. Then the local model updates are verified by all the clients and the global model will be aggregated. After that, the blockchain will create a new transaction for the verified model updates and stores it in the distributed ledger. Finally, rewards will be distributed to the deserved participants based on an incentive mechanism. Security and privacy analysis are conducted in the healthcare consortium where all the medical data, training data, local model gradients and global model gradients are distributed to all nodes' ledger. Since the consortium comprises of trusted parties and authorized actors for data access, possibilities of privacy leakages will be very rare. However, a detailed analysis in this phase would ensure to mitigate the privacy infringement risks.

Figure 7.2. Federated Learning architecture



8. Requirements Resources

8.1 Hardware Requirements

- Intel(R) Core(TM) i7 CPU, 8 GB RAM, 64-bit Ubuntu System
- Amazon Web Service (AWS) cloud platform
- AWS Elastic Kubernetes Service (EKS) Cluster - 2
- AWS EKS Node – 2 vCPU, 8 GB RAM, m5.xlarge – 1 Nos
- AWS EKS Node – 4 vCPU, 16 GB RAM, m5a.xlarge – 1 Nos
- AWS GPU instance – 4 vCPU, 61 GB RAM, p2.xlarge – 1 Nos
- AWS Elastic File System (EFS)
- AWS Elastic Compute Cloud (EC2) instance – 2 vCPU, 8 GB RAM, m.xlarge – 1 Nos
- AWS Simple Storage Service (S3)

8.2 Software Requirements

- Visual Studio Code
- Lens IDE
- JupyterLab / PyCharm

9. Research Plan

The following Fig 9.1. represents the project planner for this research

Figure 9.1. Project Planner

Blockchain-Driven Federated Learning for Healthcare Data and Image Analysis

Project Planner Select a period to highlight at right. A legend describing the charting follows.



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